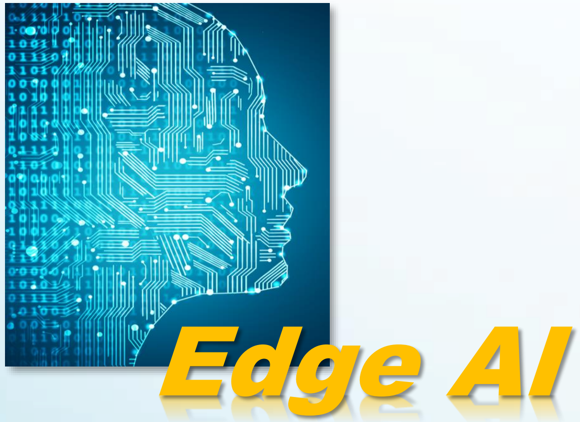
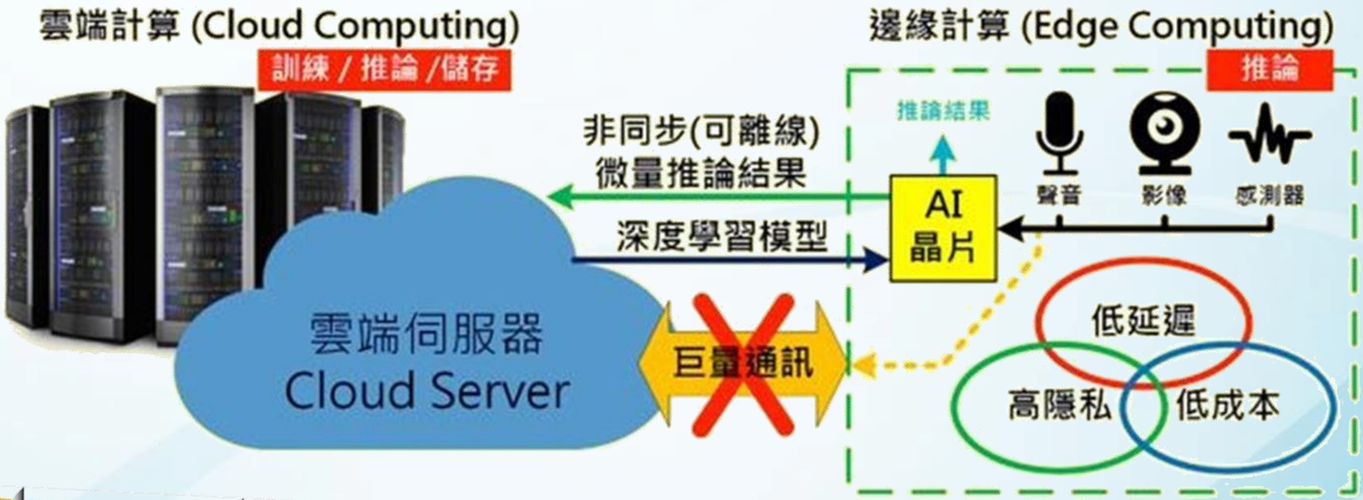
國立臺灣科技大學
NATIONAL TAIWAN UNIVERSITY OF SCIENCE AND TECHNOLOGY

邊緣人工智慧實務（EE5354701）

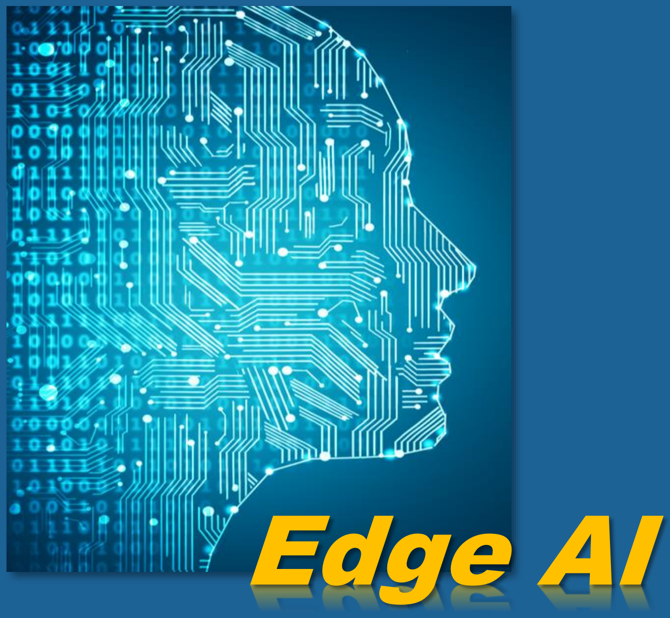




9 TinyML 案例實作 – 動作辨識

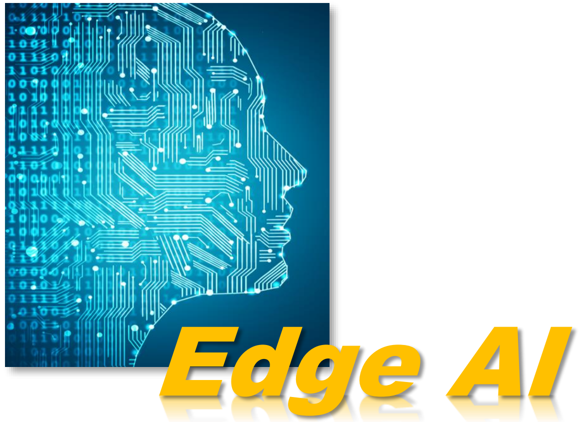
電機工程系 人工智慧應用產碩專班 許哲豪 兼任助理教授

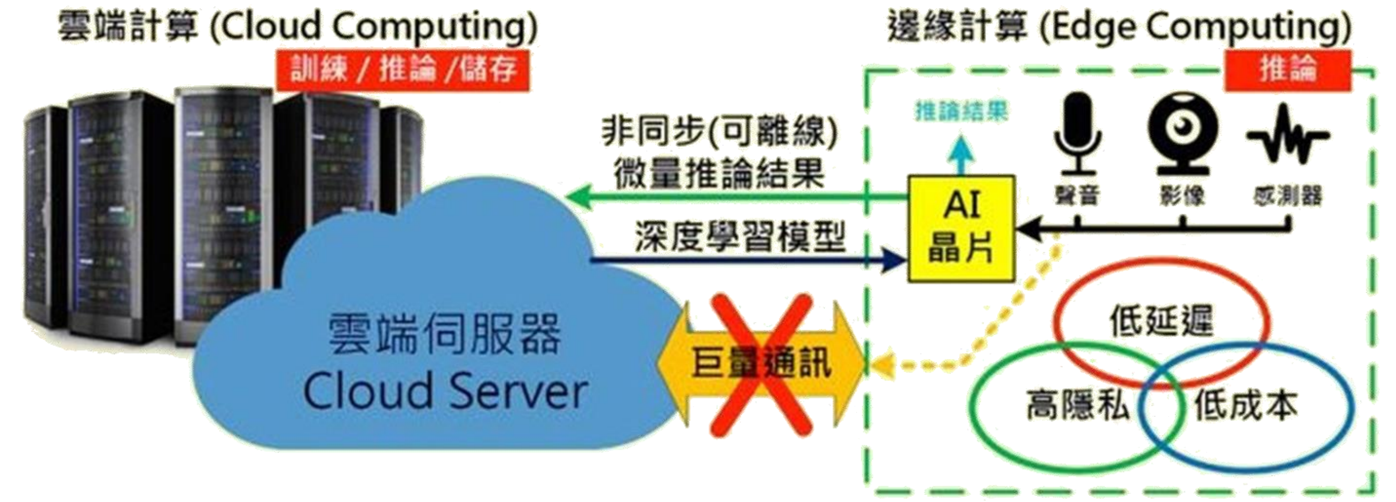
簡報大綱

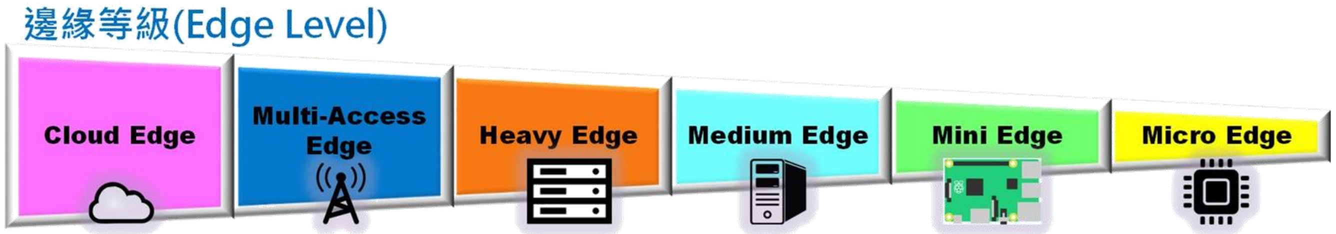


- 9.1. 運動感測器原理
- 9.2. 運動手勢辨識
- 9.3. 異常振動偵測

https://hackmd.io/@OmniXRI-Jack/NTUST_EdgeAI_2026

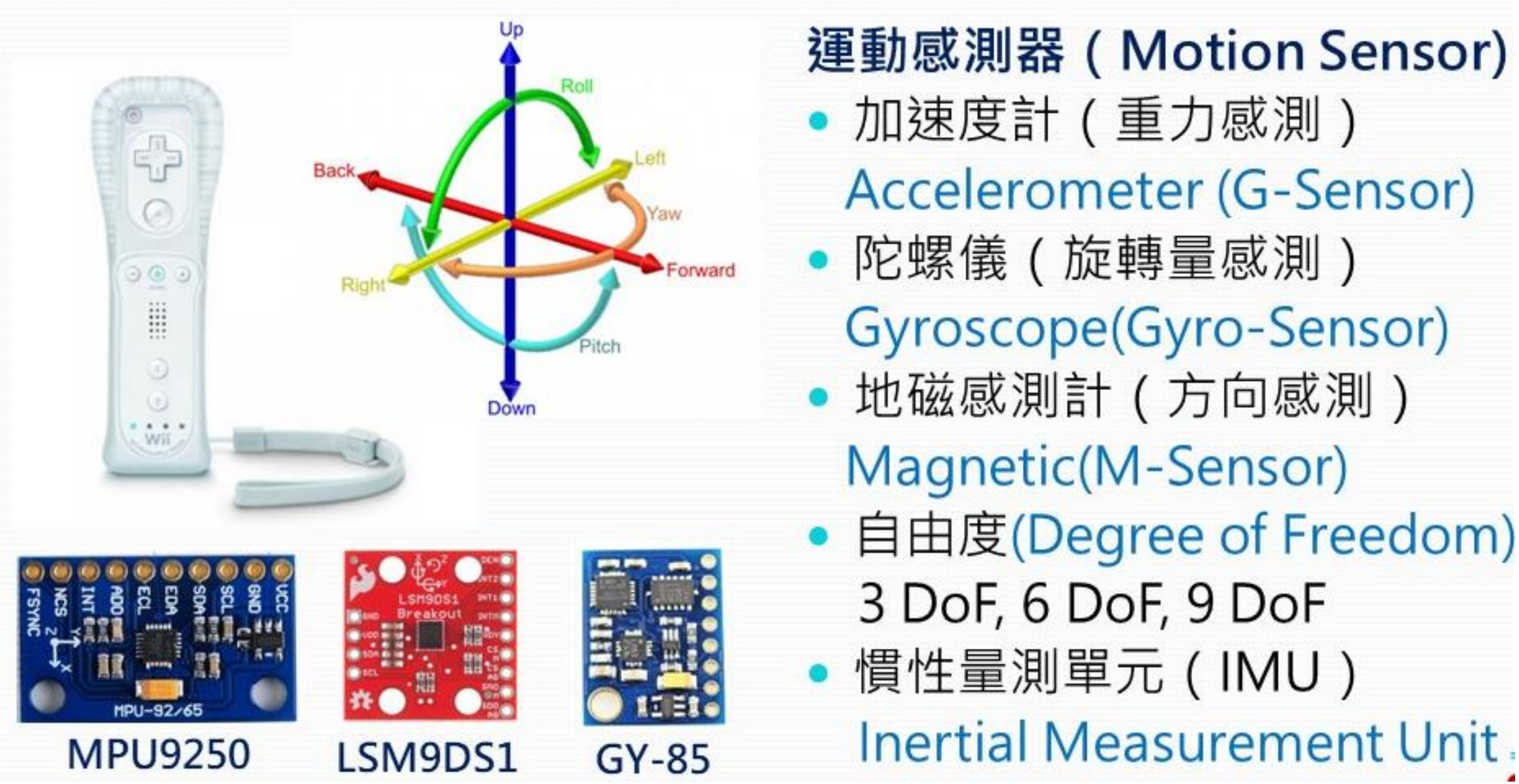

Edge AI


雲端計算 (Cloud Computing)
訓練 / 推論 /儲存
邊緣計算 (Edge Computing)
推論
非同步(可離線)
微量推論結果
深度學習模型
推論結果
AI
晶片
聲音
影像
感測器
雲端伺服器
Cloud Server
巨量通訊
低延遲
高隱私
低成本


邊緣等級(Edge Level)
Cloud Edge
Multi-Access Edge
Heavy Edge
Medium Edge
Mini Edge
Micro Edge

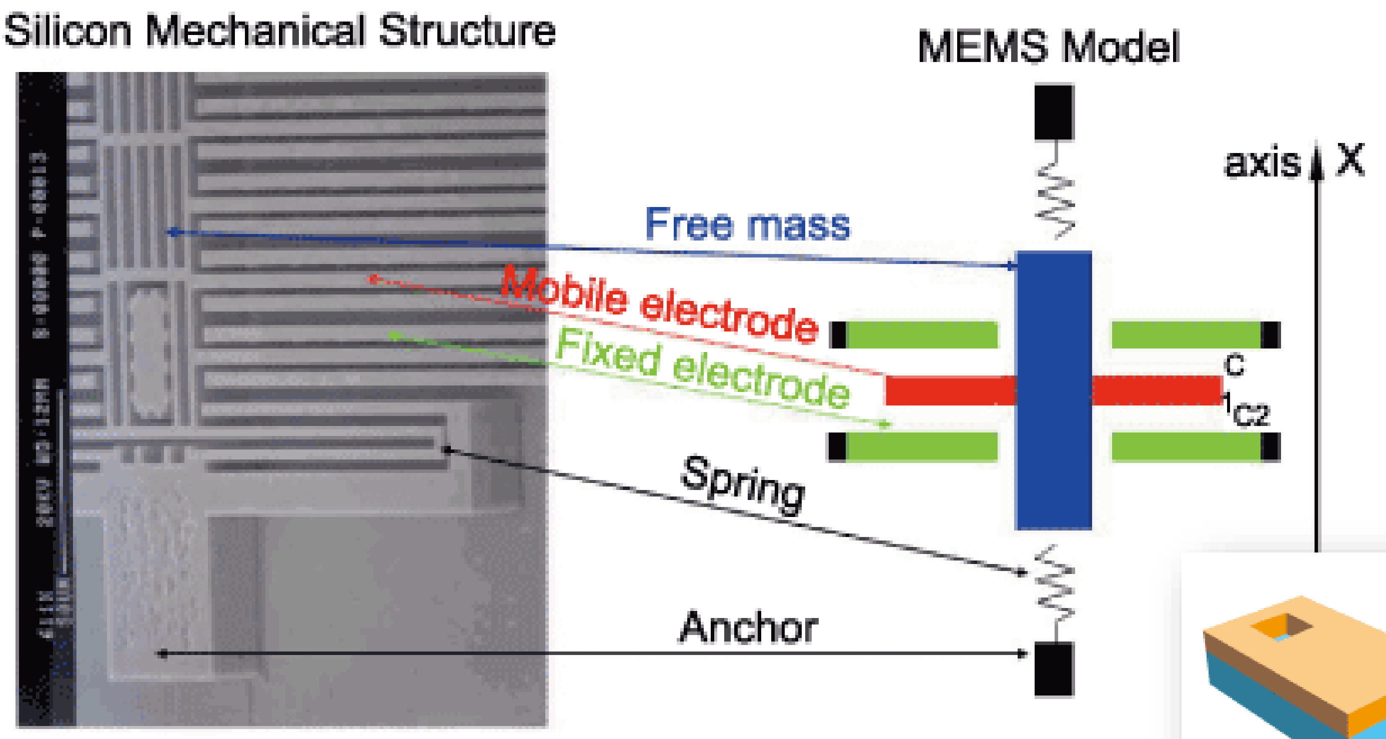
9.1. 運動感測器原理

何謂運動感測器 (Motion Sensor)

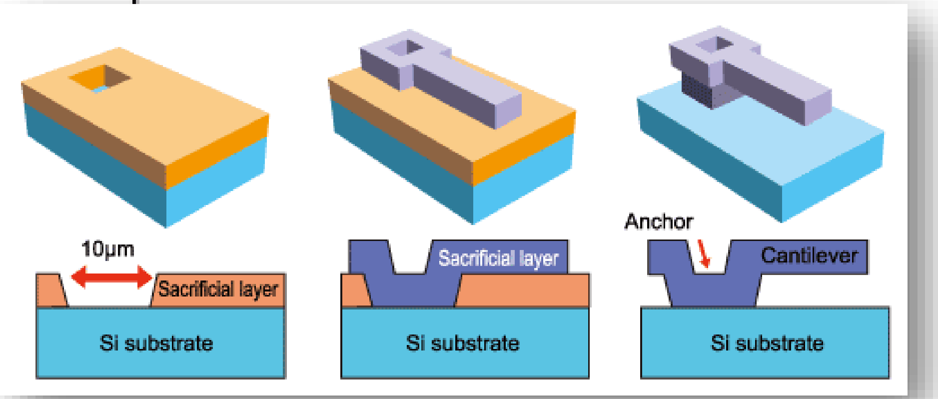


資料來源：https://omnixri.blogspot.com/2022/04/20220408.html

加速度計 (G Sensor)

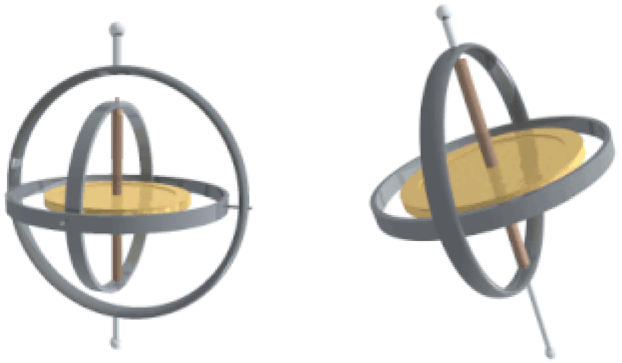


微機電MEMS
製程示意圖



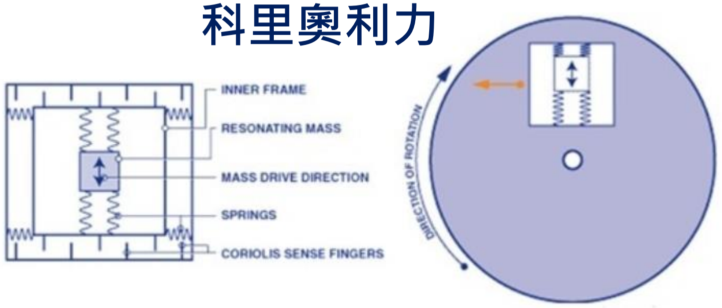
資料來源： https://www.ctimes.com.tw/DispArt/tw/%E7%BE%A9%E6%B3%95%E5%8D%8A%E5%B0%8E%E9%AB%94/MEMS/ST::%E5%8D%8A%E5%B0%8E%E9%AB%94/0711220000AW.shtml

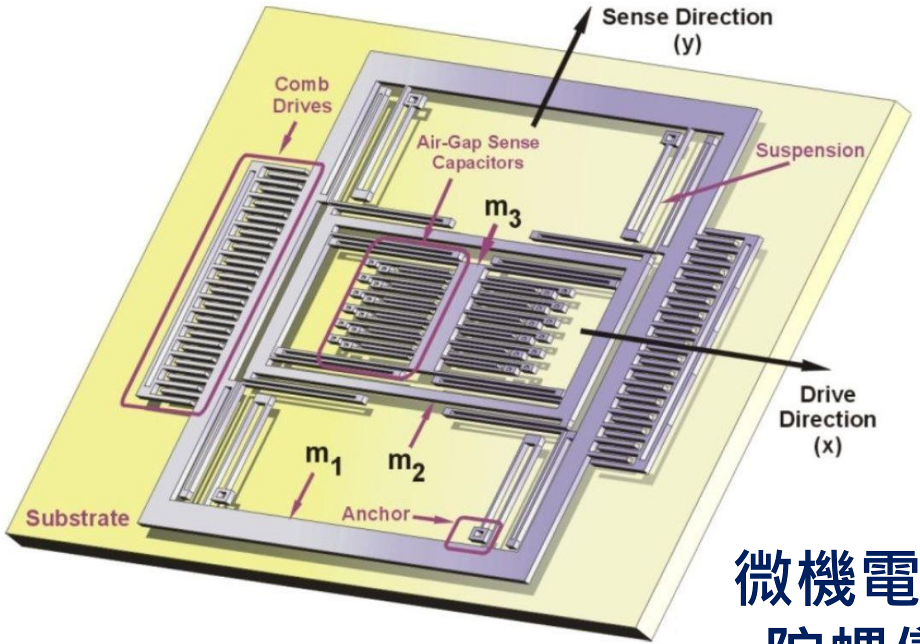
陀螺儀 (Gyro Sensor)



機械式陀螺儀

科里奧利力

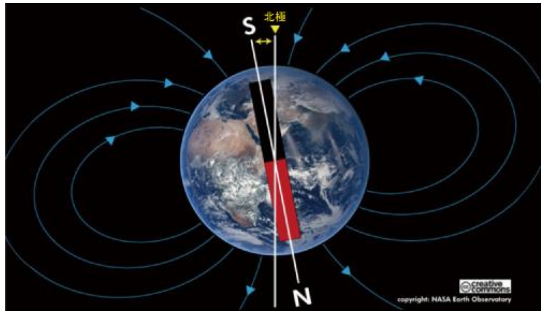


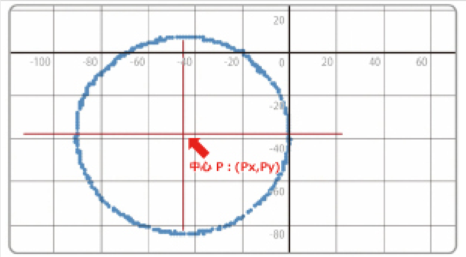


微機電式
陀螺儀

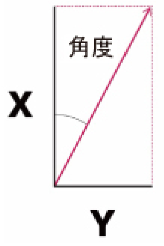
資料來源：https://zh.wikipedia.org/wiki/%E9%99%80%E8%9E%BA%E5%84%80

地磁感測器 (Geomagnetic Sensor)



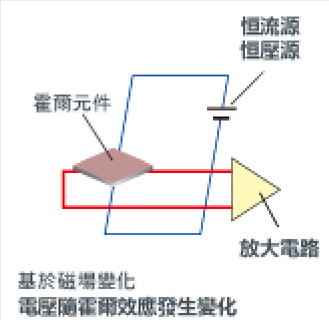
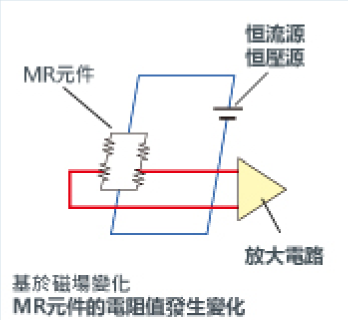
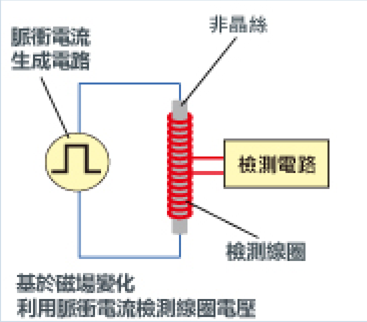


地磁感測器輸出分佈



$$角度 = \tan^{-1}\frac{Y-Py}{X-Px}$$

磁北角的計算公式

檢測方法	霍爾(Hall)	磁阻(MR)	磁抗(MI)
構成			
抗雜訊（靈敏度）	x	△	◎
消耗電流	x	△	◎
回應速度	x	△	◎

資料來源：https://www.rohm.com.tw/electronics-basics/sensors/sensor_what2

運動感測常見應用

人機互動
(Wii / PS Move 搖桿)

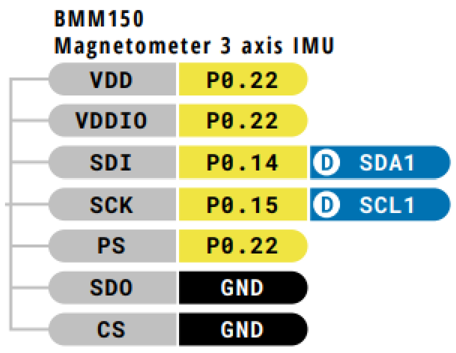
競速分析 / 駕駛習慣
(打龜號 / 車涯 04事件)

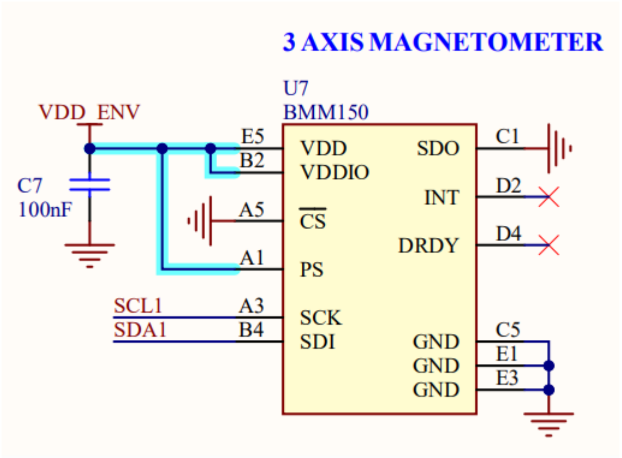
慣性導航 / 室內定位
(GPS無法看見天空)

異常振動
(馬達/ 輸送帶)

自動平衡
(平衡車、無人機、人形機器人)

Arduino Nano 33 BLE Sense Rev.2





BMM150

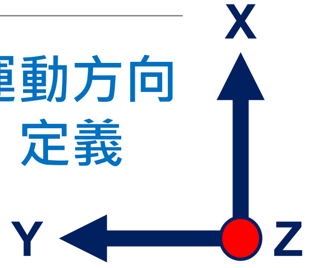
3軸地磁計

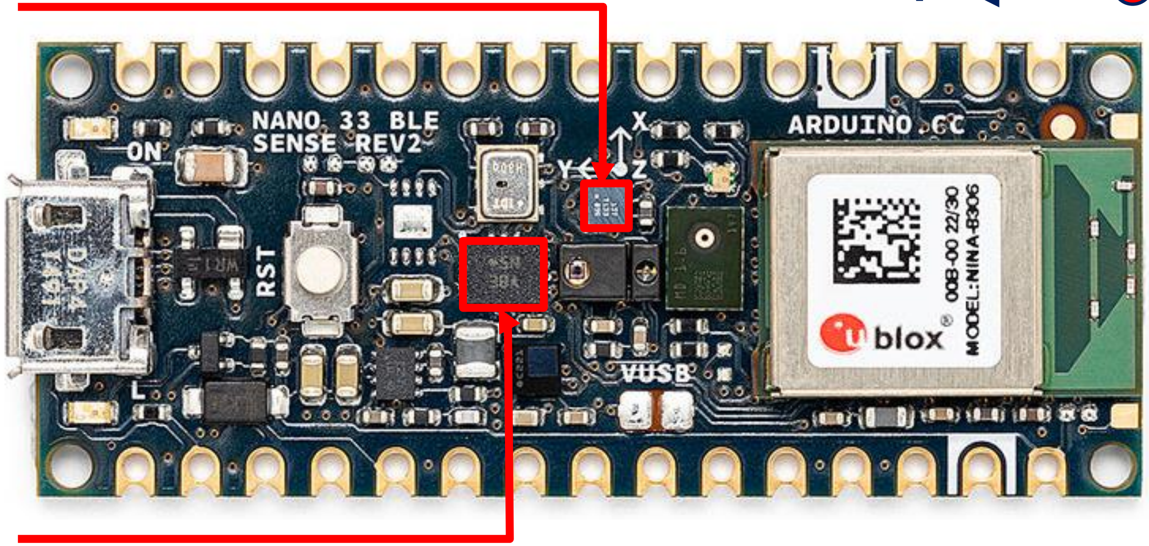
±1300μT (x, y-axis),

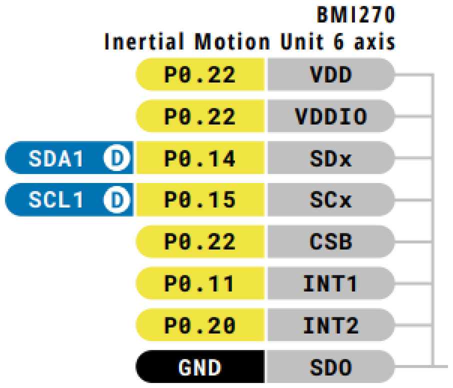
±2500μT (z-axis) @16bit

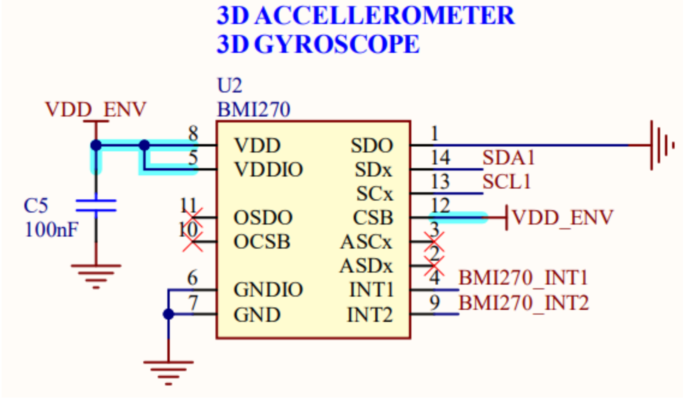
ODR 2Hz ~ 30 Hz

運動方向定義









BMI270

3軸加速度計 ±2g/±4g/±8g/±16g @16bit

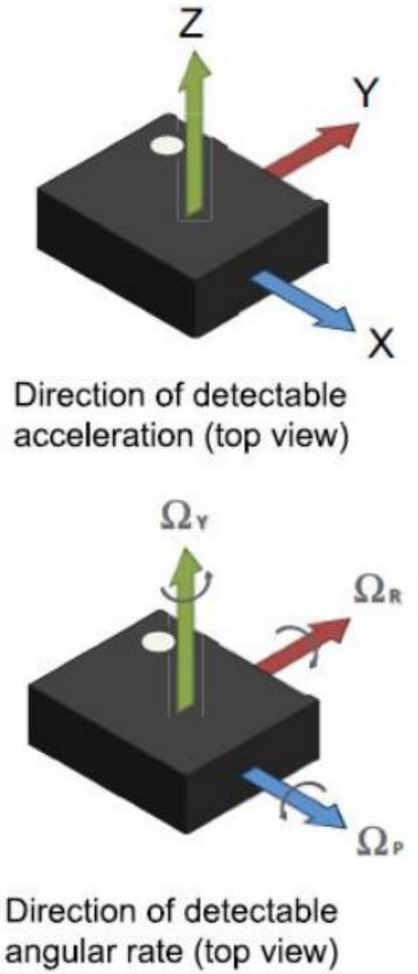
ODR 0.78Hz ~ 1.6 kHz

3軸陀螺儀 ±125/±250/±500/±1K/±2Kdps @16bit

ODR 25 Hz ~ 6.4 kHz

資料來源：https://docs.arduino.cc/hardware/nano-33-ble-sense-rev2/

Seeed Xiao nRF52840 Sense

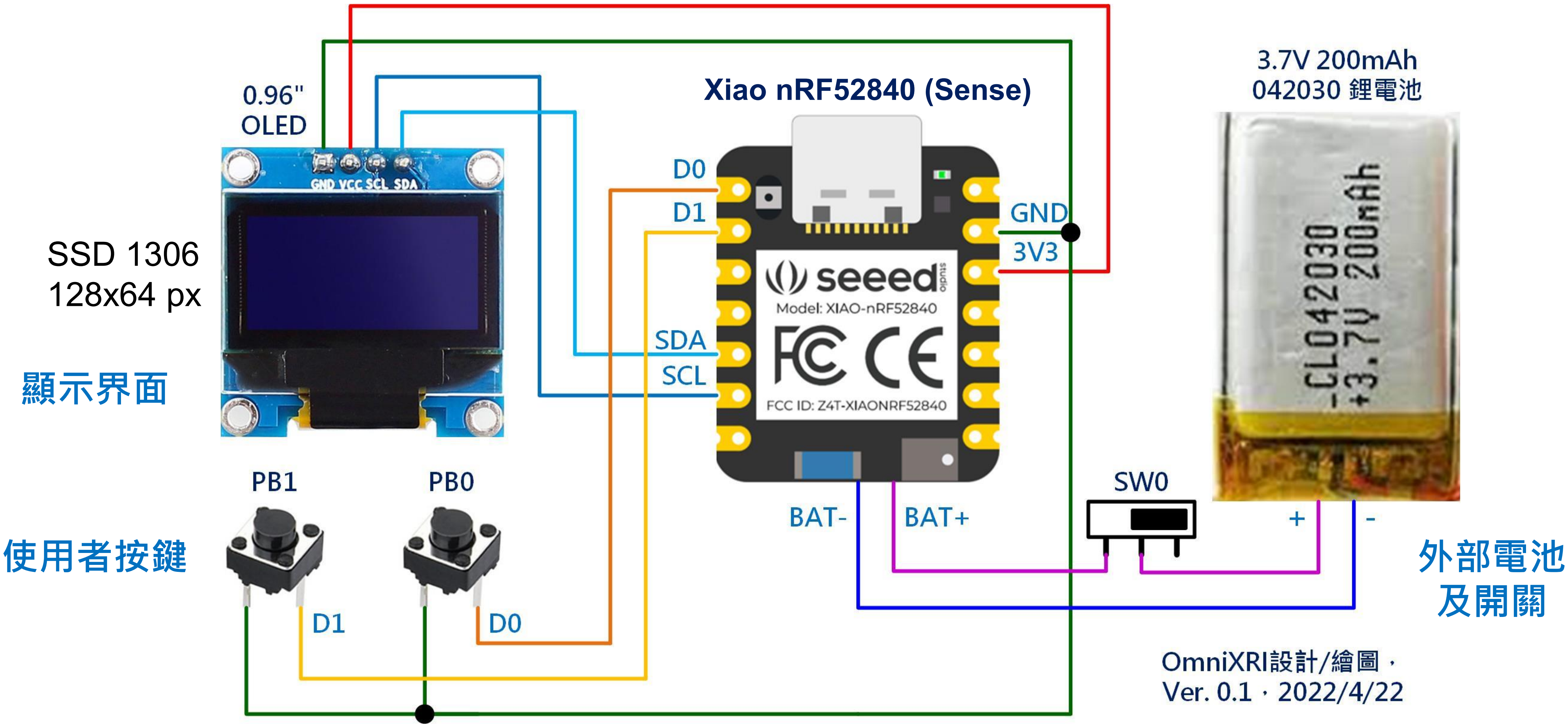


https://wiki.seeedstudio.com/cn/XIAO_BLE/

LSM6DS3TR-C（IMU）

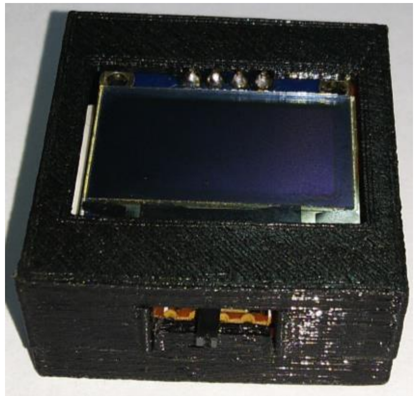
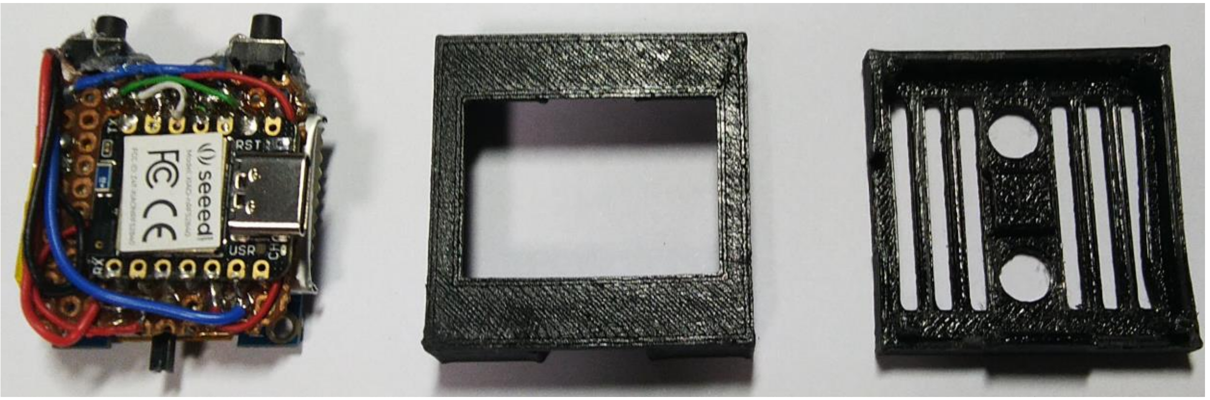
3軸加速度計	±2g/±4g/±8g/±16g @16bit ODR 12.5Hz ~ 1.66KHz
3軸陀螺儀	±125/±250/±500/±1K/±2Kdps @16bit ODR 12.5Hz ~ 1.66KHz
溫度感測器	-40 ~ +85 ℃ @12bit ODR 52Hz

穿戴式智慧人工智慧裝置 – 參考電路



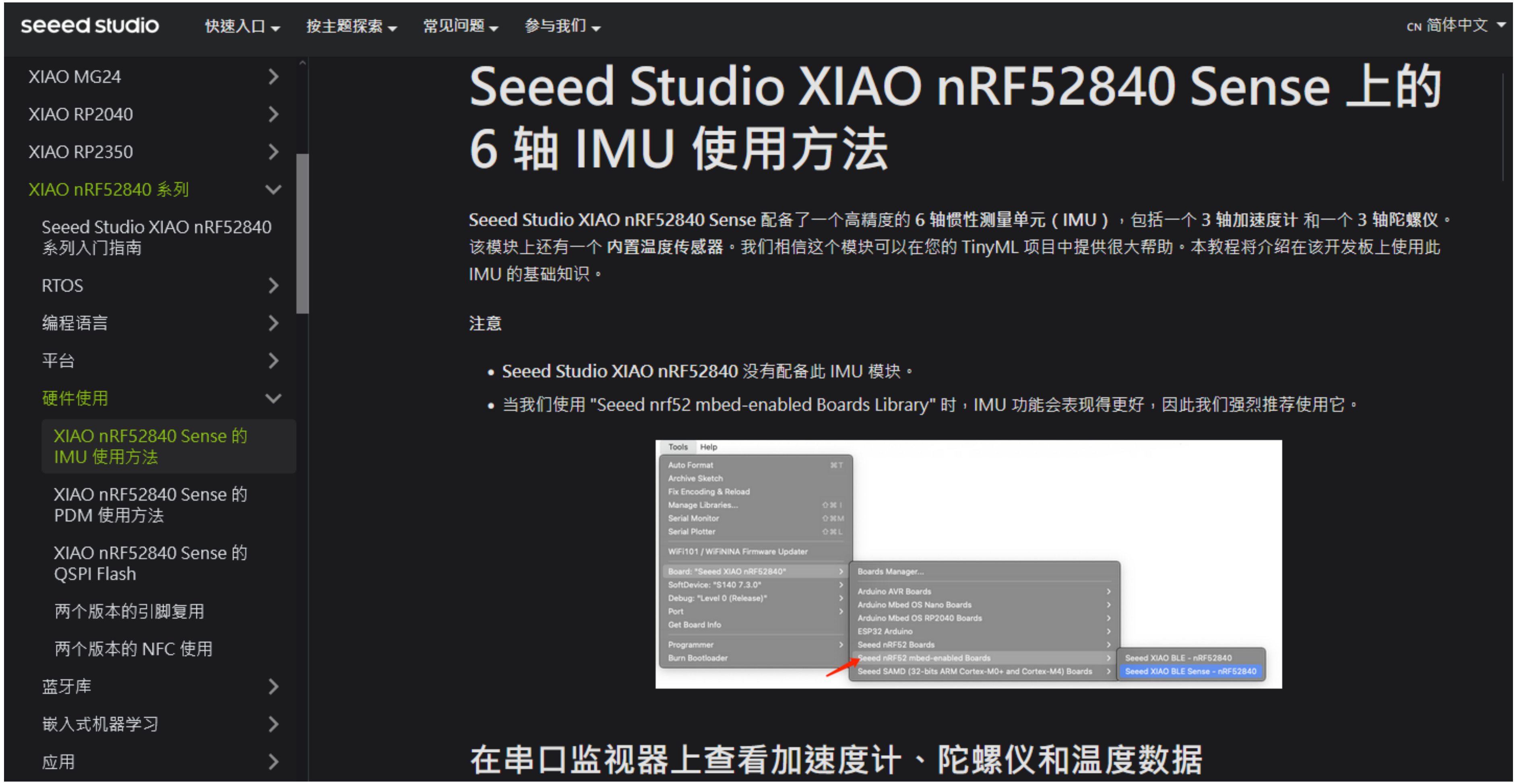
穿戴式智慧人工智慧裝置 – 參考外形



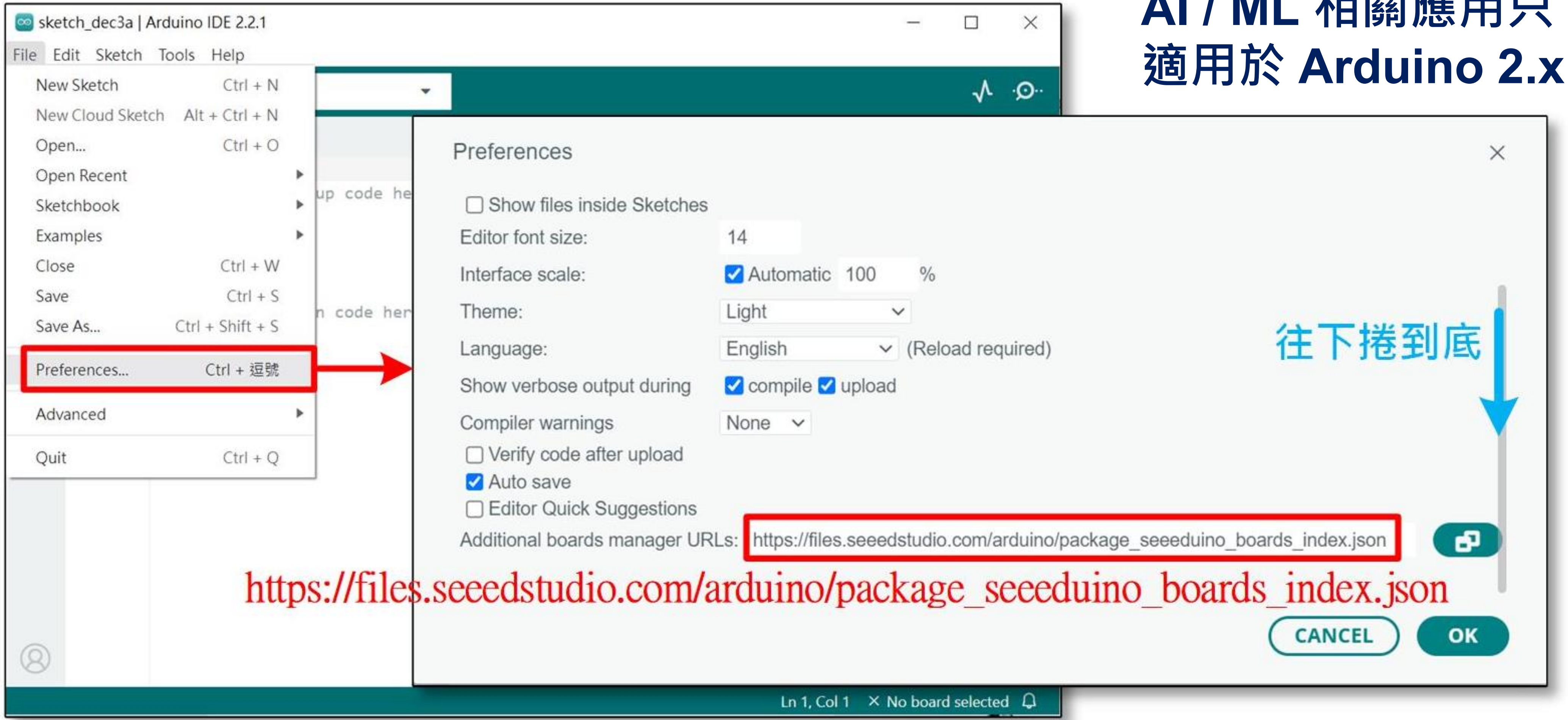

資料來源：https://youtu.be/HxnjNGnKn2Y?list=PLo_tFZGoEZum8b35jClphx5kL8AP1bGut

Xiao nRF52840 Sense IMU使用方法



https://wiki.seeedstudio.com/cn/XIAO-BLE-Sense-IMU-Usage/

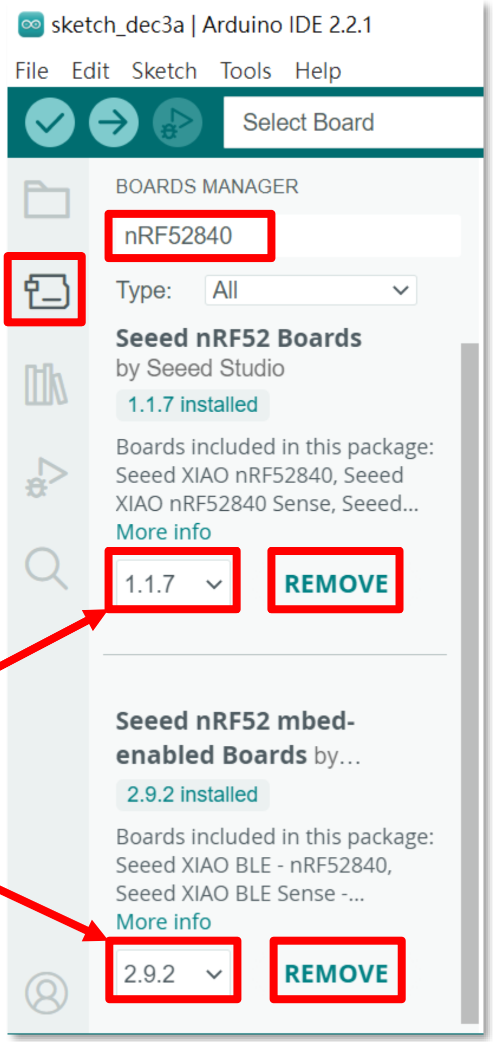
Arduino 新增開發板設定



AI / ML 相關應用只適用於 Arduino 2.x。

OmniXRI整理製作, 2023/12/08

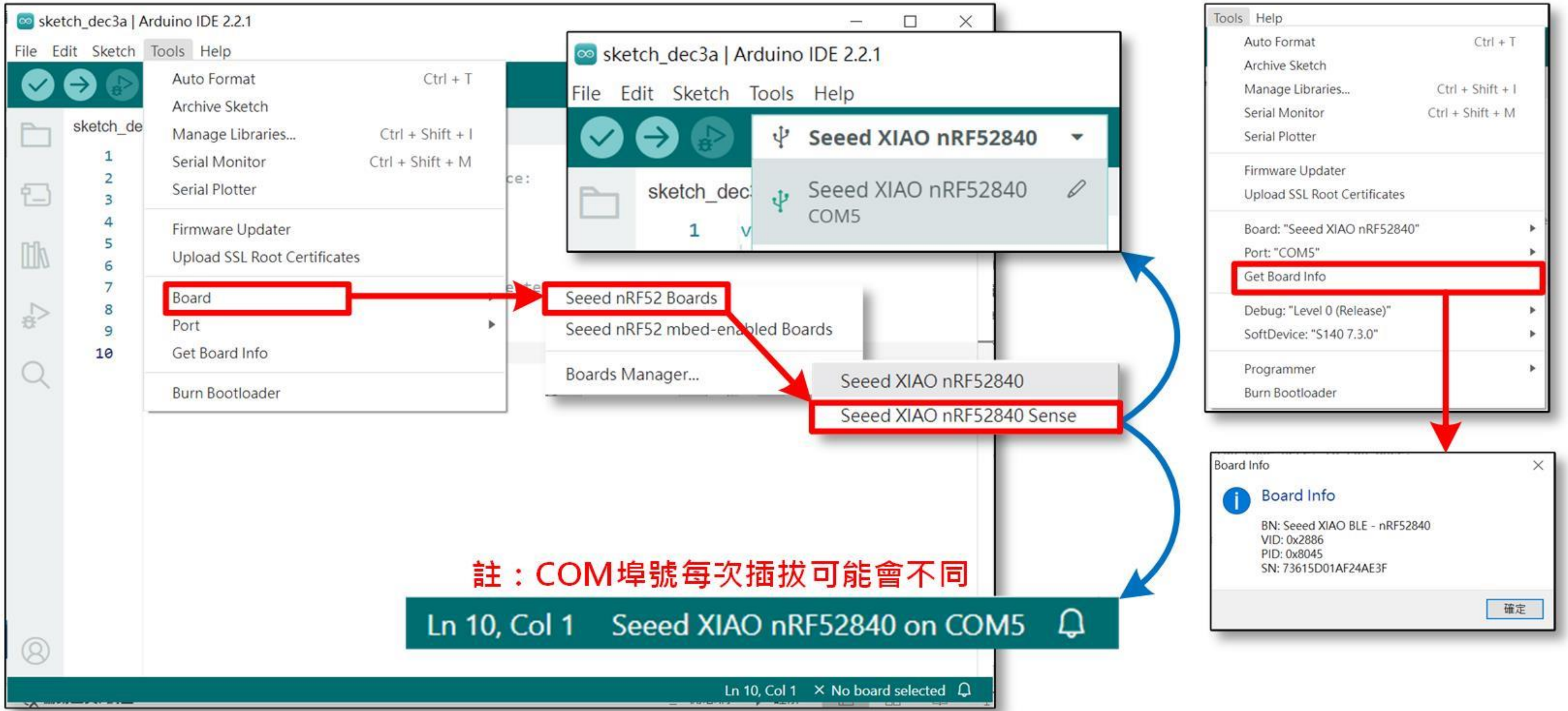
安裝Seeed nRF52840函式庫



版本選最新的

- 點選選單 Tools > Board > Boards Manager... ，或直接點選左側第二個「開發板」圖示。
- 輸入 nRF52840 搜尋 Seeed nRF52840 開發板相關函式庫。
- 點選「INSTALL」安裝下列二個函式庫。（版本可取最新的）
  - Seeed nRF52 Boards（BLE, 低功耗功能）
  - Seeed nRF52 mbed-enabled Boards（PDM, IMU, ML）
- 安裝後若不需要時，可點選「REMOVE」解除安裝。

指定工作開發板及埠號

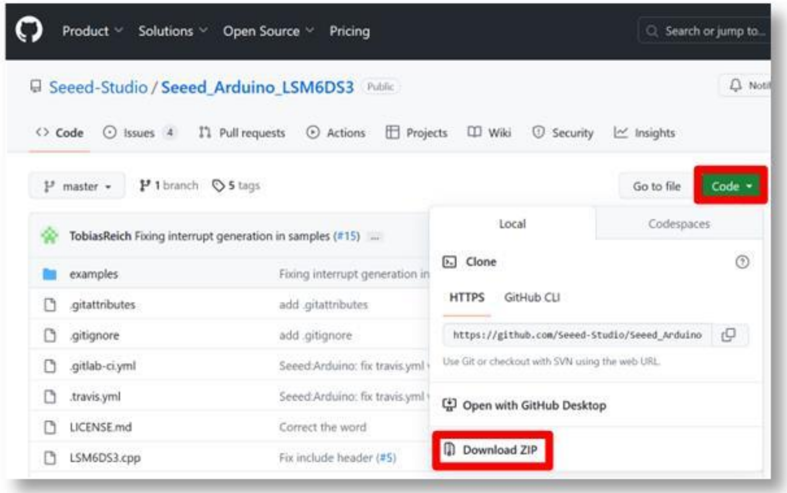


OmniXRI整理製作, 2023/12/08

- 選擇開發板 Seeed XIAO nRF52840 Sense
- 選擇對應埠號
- 檢查是否連線

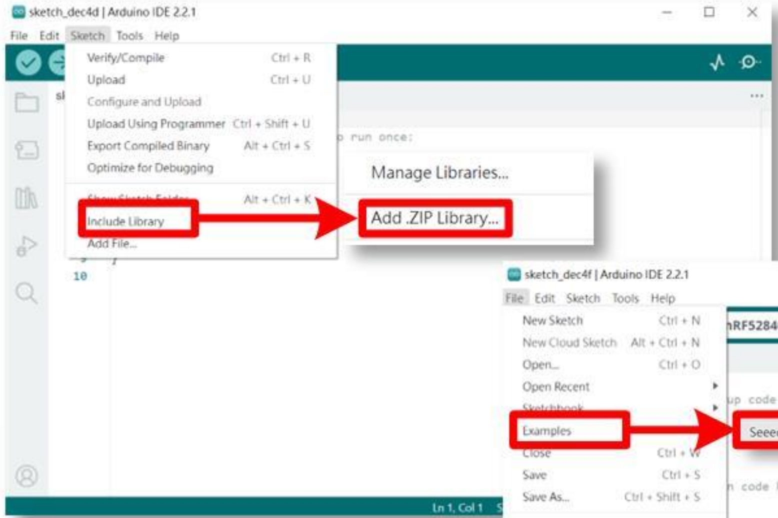
安裝運動感測器（IMU）函式庫

https://github.com/Seeed-Studio/Seeed_Arduino_LSM6DS3



Seeed_Arduino_LSM6DS3-master.zip

下載運動感測器函式庫



安裝函式庫

開啟範例程式

https://github.com/Seeed-Studio/Seeed_Arduino_LSM6DS3

OmniXRI整理製作, 2023/12/08

運動感測器（IMU）取值範例

國立臺灣科技大學
NATIONAL TAIWAN UNIVERSITY OF SCIENCE AND TECHNOLOGY

File > Examples > Seeed Aruino LSM6DS3 > IMU_capture

```
#include <LSM6DS3.h>
#include <Wire.h>
LSM6DS3 myIMU(I2C_MODE, 0x6A); // 建立LSM6DS3實例並設定I2C裝置位址
float aX, aY, aZ, gX, gY, gZ; // 儲存三軸加速度及陀螺儀數值變數
const float accelerationThreshold = 2.5; // 加速度門檻值
const int numSamples = 119; // 樣本數量
int samplesRead = numSamples; // 讀取樣本數量
void setup() { // 初始化設定
  Serial.begin(9600); // 設定串例通信速度(bps)
  while (!Serial); // 等待串列埠初始化完成
  if (myIMU.begin() != 0) { // 初始化運動感測器
    Serial.println("Device error");
  } else {
    Serial.println("aX,aY,aZ,gX,gY,gZ");
  }
}
void loop() { // 無限迴圈
  // 等待取得足夠數量樣本
  while (samplesRead == numSamples) {
    // 讀取加速度計值
    aX = myIMU.readFloatAccelX();
    aY = myIMU.readFloatAccelY();
    aZ = myIMU.readFloatAccelZ();
    // 求得加速度值總合
    float aSum = fabs(aX) + fabs(aY) + fabs(aZ);
    // 檢查是否超過門檻值，若是重置已讀樣本數量
    if (aSum >= accelerationThreshold) {
      samplesRead = 0;
      break;
    }
  }
```

觸發值大於門檻才輸出

```
  // 檢查自上次檢測到顯著運動以來是否已讀取所有必需的樣本
  while (samplesRead < numSamples) {
    samplesRead++; // 已讀取樣本數加1
    // print the data in CSV format
    Serial.print(myIMU.readFloatAccelX(), 3);
    Serial.print(',');
    Serial.print(myIMU.readFloatAccelY(), 3);
    Serial.print(',');
    Serial.print(myIMU.readFloatAccelZ(), 3);
    Serial.print(',');
    Serial.print(myIMU.readFloatGyroX(), 3);
    Serial.print(',');
    Serial.print(myIMU.readFloatGyroY(), 3);
    Serial.print(',');
    Serial.print(myIMU.readFloatGyroZ(), 3);
    Serial.println();
    // 若已讀樣本數已足夠則列印換行
    if (samplesRead == numSamples) {
      Serial.println();
    }
  }
}
```

練習：將板子以不同方向移動及旋轉測試讀值是否正常

IMU 連續取值輸出至 Edge Impulse

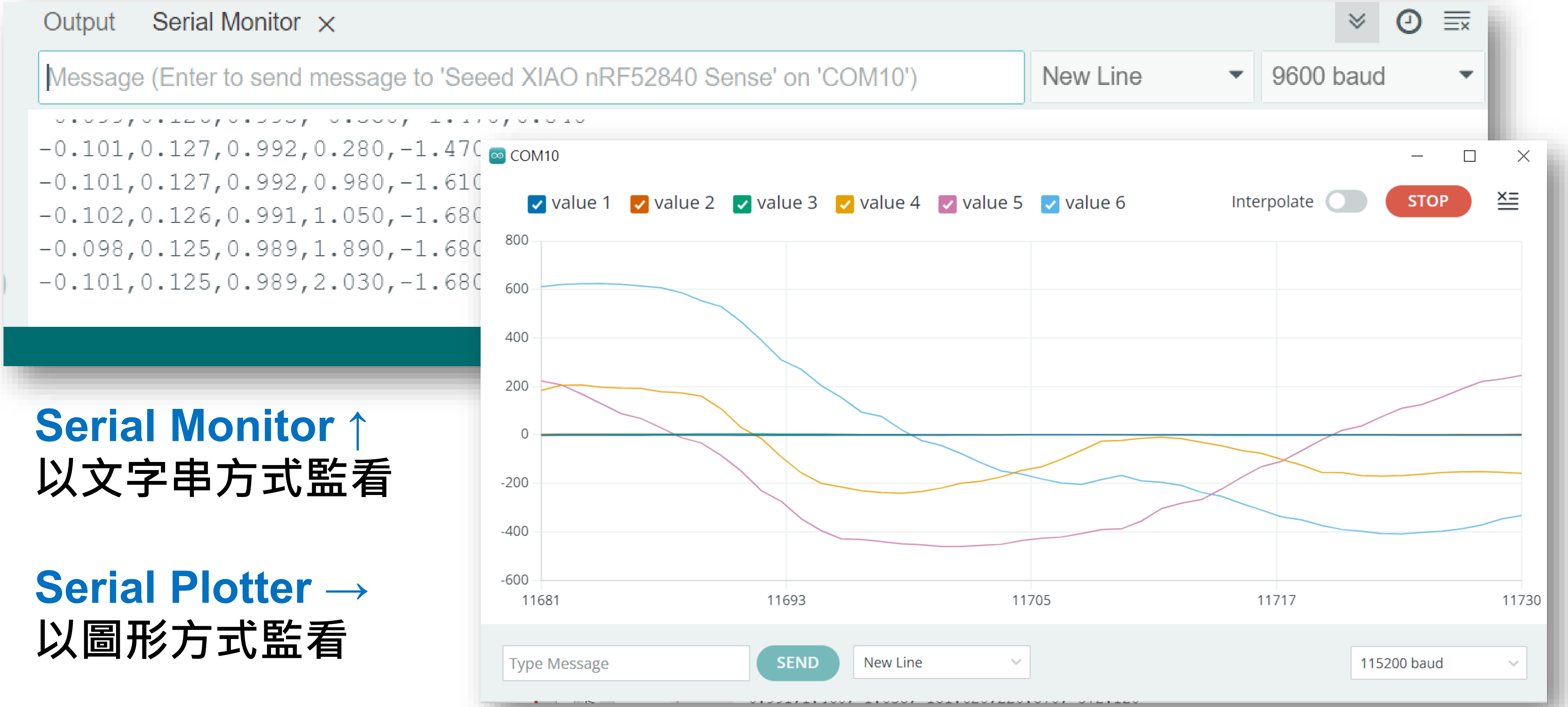
```
#include <LSM6DS3.h>
#include <Wire.h>

//Create a instance of class LSM6DS3
LSM6DS3 myIMU(I2C_MODE, 0x6A);    //I2C device address 0x6A
float aX, aY, aZ, gX, gY, gZ;
// const float accelerationThreshold = 2.5; // threshold of significant in G's
// const int numSamples = 119;
// int samplesRead = numSamples;

void setup() {
  // put your setup code here, to run once:
  Serial.begin(115200);
  while (!Serial);
  //Call .begin() to configure the IMUs
  if (myIMU.begin() != 0) {
    Serial.println("Device error");
  } else {
    Serial.println("aX,aY,aZ,gX,gY,gZ");
  }
}

void loop() {
  // wait for significant motion
//   while (samplesRead == numSamples) {
//     // read the acceleration data
//     aX = myIMU.readFloatAccelX();
//     aY = myIMU.readFloatAccelY();
//     aZ = myIMU.readFloatAccelZ();

//     // sum up the absolutes
//     float aSum = fabs(aX) + fabs(aY) + fabs(aZ);

//     // check if it's above the threshold
```

註解或刪除
紅色框區域

```
//      if (aSum >= accelerationThreshold) {
//        // reset the sample read count
//        samplesRead = 0;
//        break;
//      }
//    }

//    // check if the all the required samples have been read since
//    // the last time the significant motion was detected
//    while (samplesRead < numSamples) {
//      // check if both new acceleration and gyroscope data is
//      // available
//      // read the acceleration and gyroscope data

//      samplesRead++;

     // print the data in CSV format
     Serial.print(myIMU.readFloatAccelX(), 3);
     Serial.print(',');
     Serial.print(myIMU.readFloatAccelY(), 3);
     Serial.print(',');
     Serial.print(myIMU.readFloatAccelZ(), 3);
     Serial.print(',');
     Serial.print(myIMU.readFloatGyroX(), 3);
     Serial.print(',');
     Serial.print(myIMU.readFloatGyroY(), 3);
     Serial.print(',');
     Serial.print(myIMU.readFloatGyroZ(), 3);
     Serial.println();

//    if (samplesRead == numSamples) {
//      // add an empty line if it's the last sample
//      Serial.println();
//    }
//  }
}
```

連續取值並輸出至 COM
搭配 **edge-impulse-data-forwarder** 可將感測器值送到雲端

監看運動感測器（IMU）輸出值



Serial Monitor ↑
以文字串方式監看

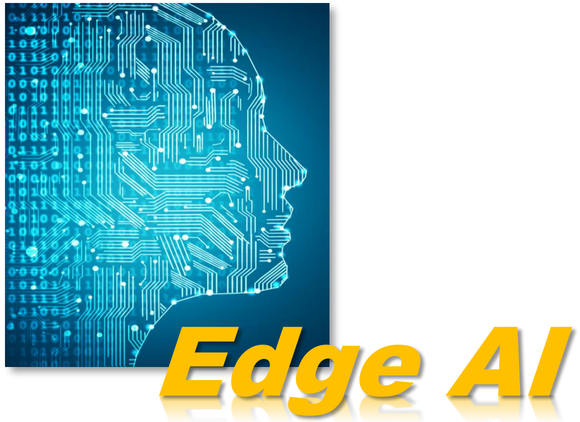
Serial Plotter →
以圖形方式監看

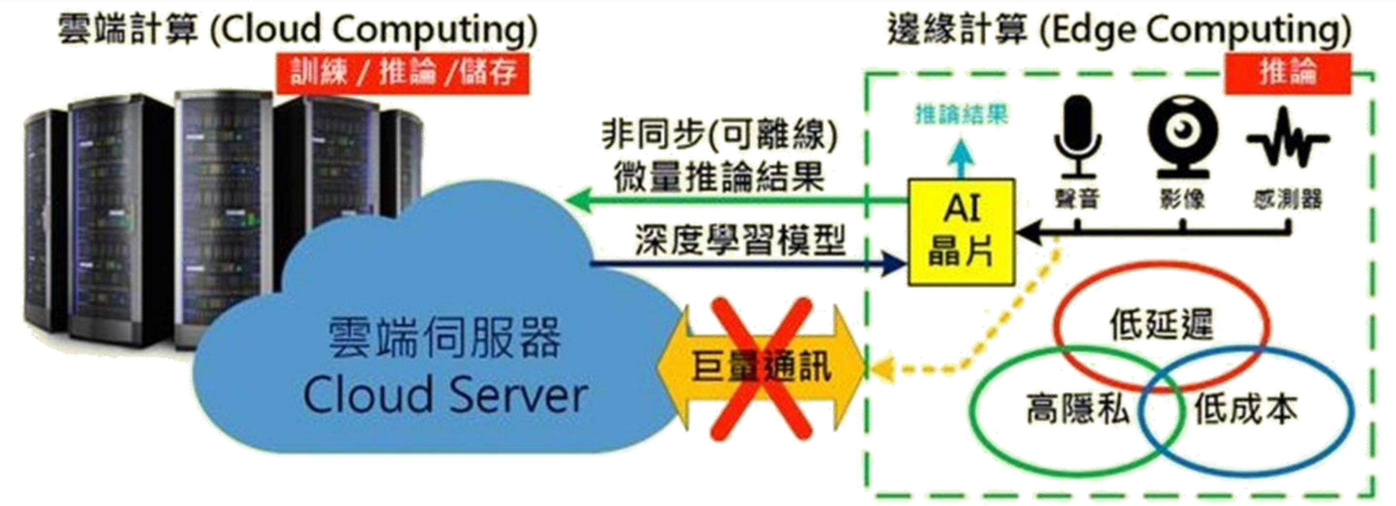
案例分享 – Rabboni

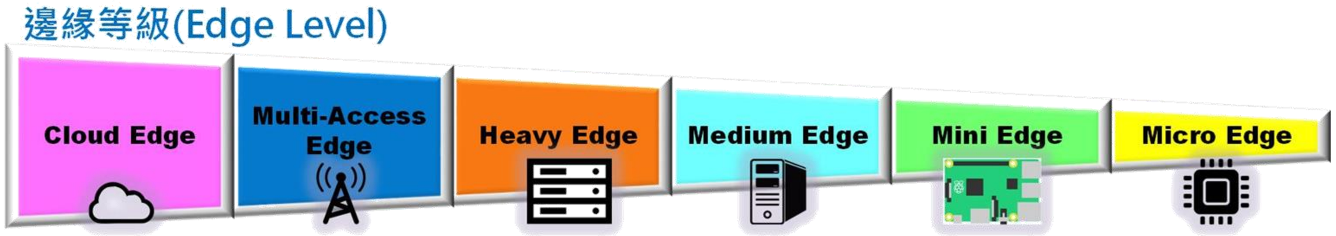
陽明交大電機工程系溫瓌岸教授團隊衍生新創公司產品，內建六軸重力感測器、藍芽傳輸及運算元件，可即時傳輸感測讀值並提供取樣頻率及動態範圍之多樣選擇。

2025 陽明交大攜手奇景光電推動AI 應用創新菁英培育計畫決賽暨成果發表會

資料來源：https://rabboni.com.tw/


Edge AI


雲端計算 (Cloud Computing)
訓練 / 推論 /儲存
邊緣計算 (Edge Computing)
推論
非同步(可離線)
微量推論結果
深度學習模型
推論結果
AI
晶片
聲音
影像
感測器
雲端伺服器
Cloud Server
巨量通訊
低延遲
高隱私
低成本

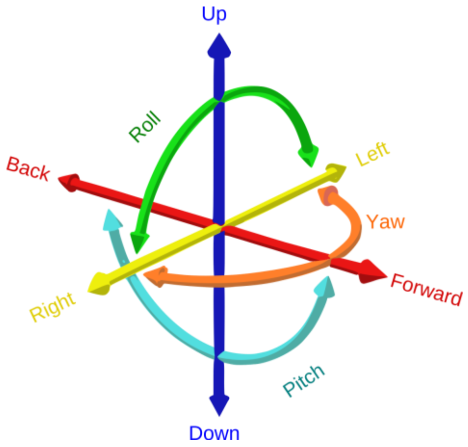

邊緣等級(Edge Level)
Cloud Edge
Multi-Access Edge
Heavy Edge
Medium Edge
Mini Edge
Micro Edge

9.2. 運動手勢辨識

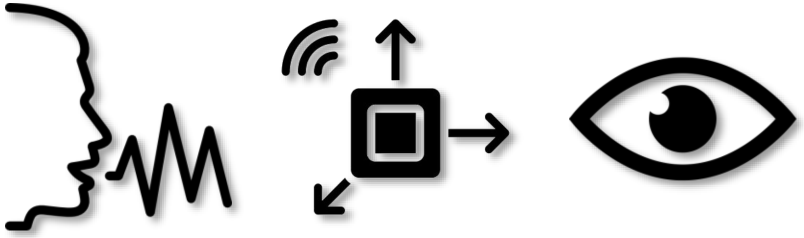
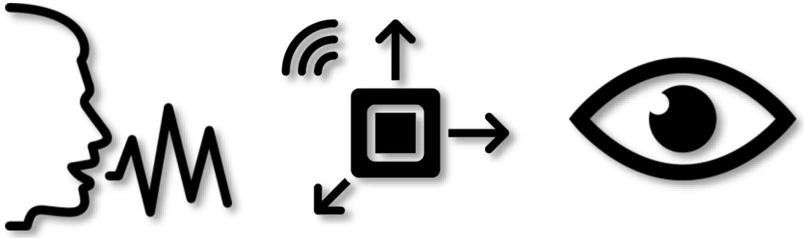
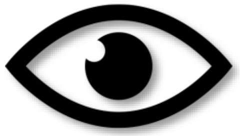
設計自定義操作手勢

- Start – 向前張開手掌
- Next – 由左向右快速平移並有頓點
- Prev – 由下向上擺動並有頓點
- Stop – 雙手快速交叉於胸口
- Unknow – 任意輕微擺動或不動，可為任意姿勢。

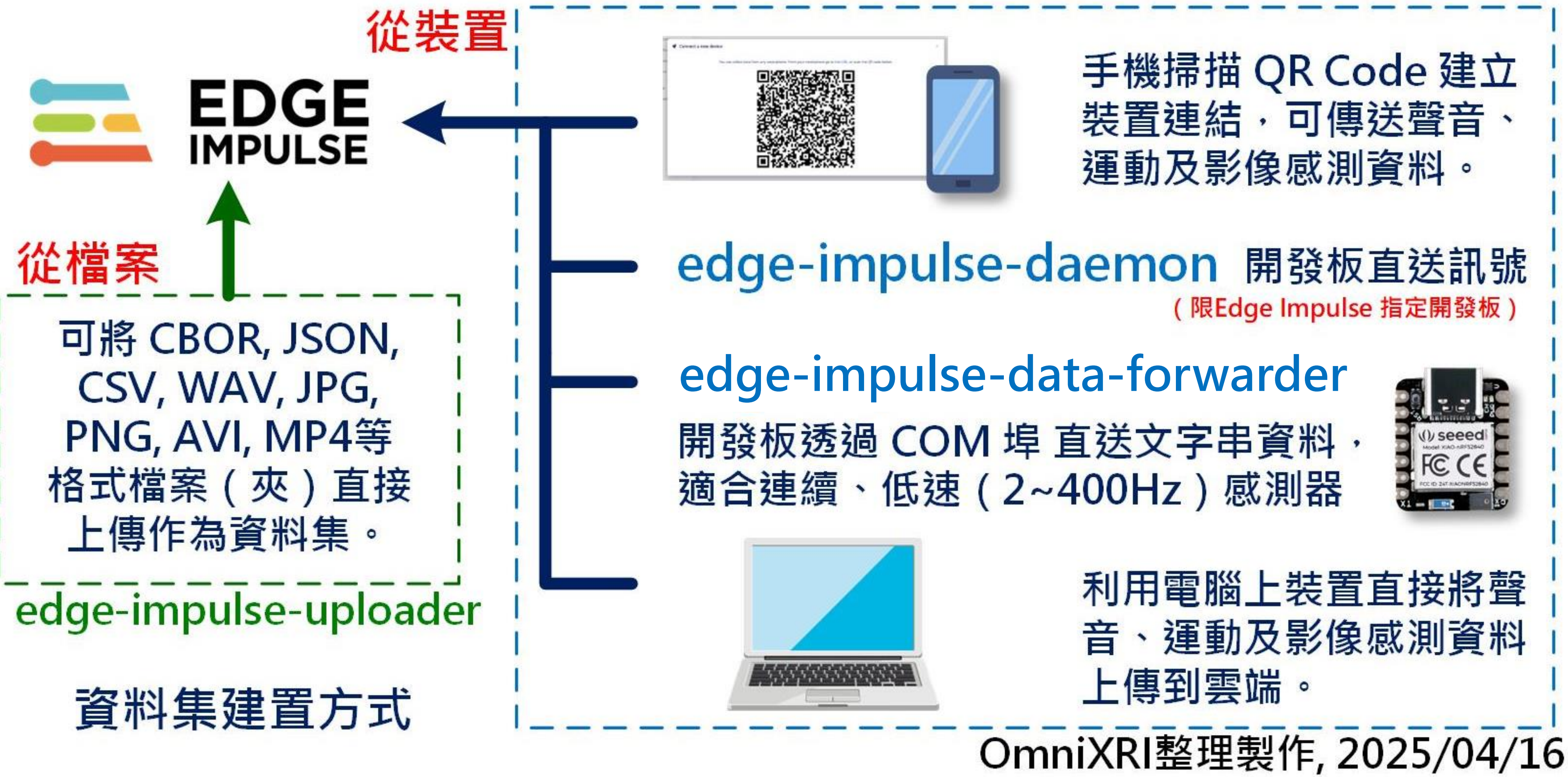
- **運動感測器必須固定在手部特定位置以免相同動作但感測值差異過大**
- **動作設定最好於不同軸向有明顯差異可只用加速度計而不使用陀螺儀**



TinyML 開發流程選項

模型訓練工具名稱	適用感測器	主要特色
TensorFlow Lite (現更名LiteRT)		免費版：資料集需自行建置，僅提供模型訓練及轉成TensorFlow Lite for Microcontroller 格式。可利用雲端 Google Colab 或本地端 Jupyter Notebooks 進行模型訓練及部署。
EDGE IMPULSE		免費版：可支援各式感測器及開發板，提供一定額度資料集建置、資料前處理（特徵提取）模型訓練、模型優化及多種部署方式。亦可導入已訓練好的 TensorFlow Lite, ONNX 等模型。
SenseCraft		免費版：影像分類可多分類、物件偵測同一模型僅能一類。事件觸發只能設一種條件。可支援雲端資料收集、模型訓練及部署，亦導入外部已訓練好之 TensorFlow Lite 模型。

Edge Impulse 資料集建置方式



快速操作指令表

https://github.com/OmniXRI/Edge_AI_TinyML_Course_2024/blob/main/Ch13_Motion_Recognition/IMU_Quick_Guide.md

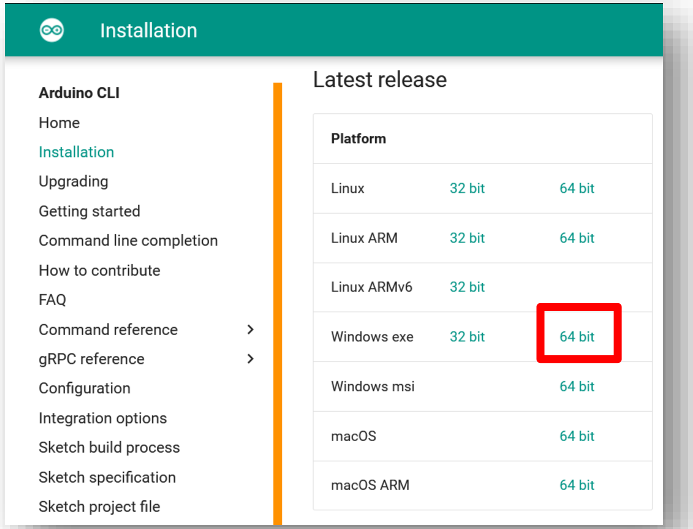
下載及安裝必要工具

國立臺灣科技大學 NATIONAL TAIWAN UNIVERSITY OF SCIENCE AND TECHNOLOGY

1. Python 3.x

3. Arduino CLI

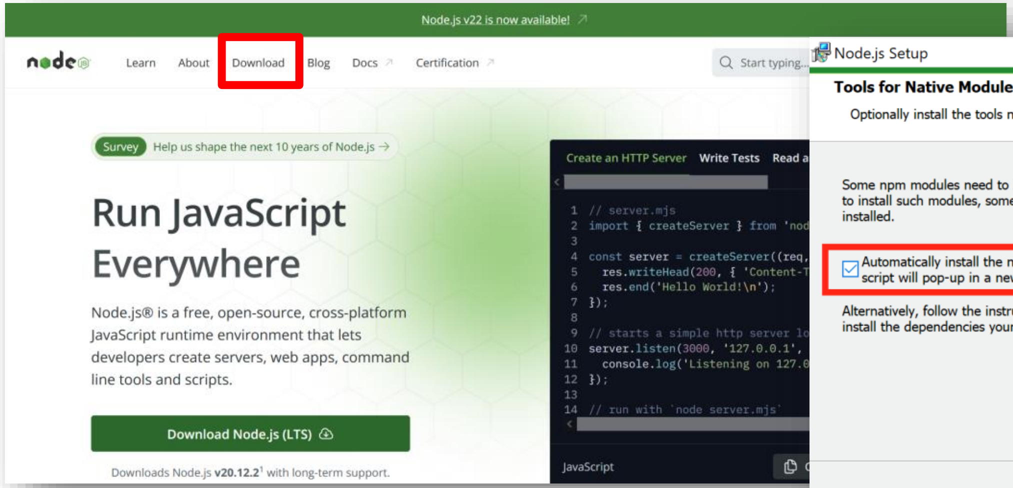
(選配，依不同開發板需求)

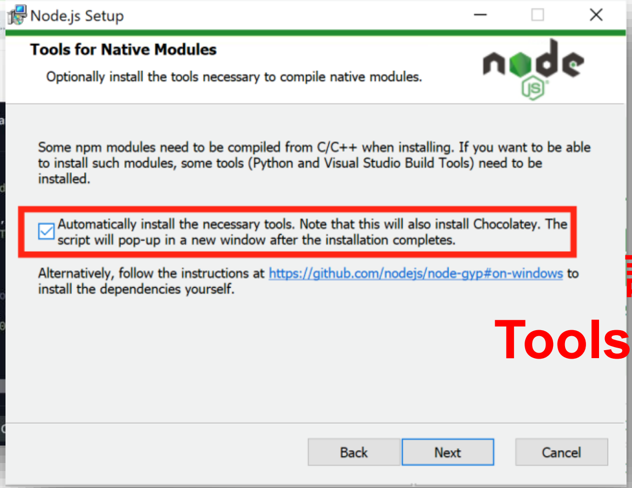


2. Node.js

(若後續 Edge Impulse 無法安裝

可換回舊版，例： v18.20.2 LTS)





安裝時要注意

記得一定要勾選

Tools for Native Modules

以 node -v 及 npm -v

檢查安裝好的版本

安裝 Edge Impulse windows 工作環境

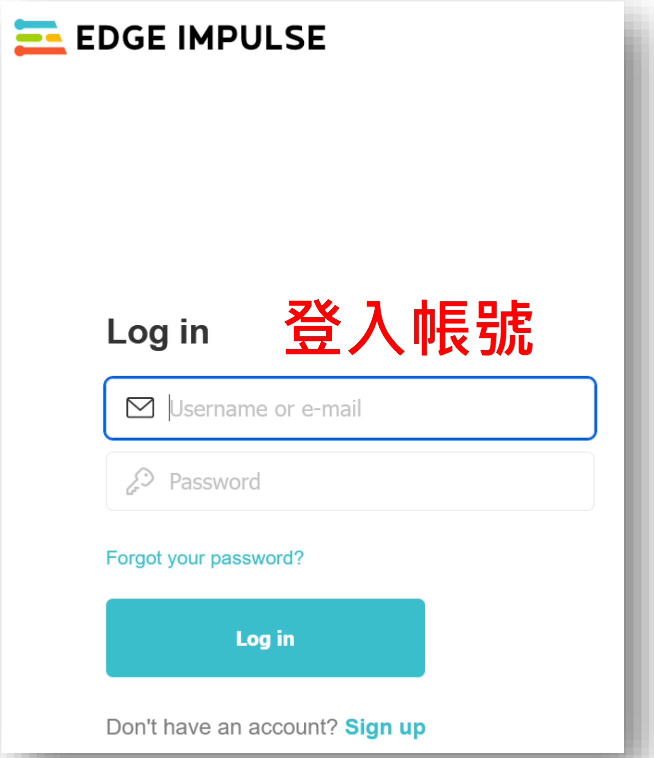
- 安裝**Edge Impulse CLI**

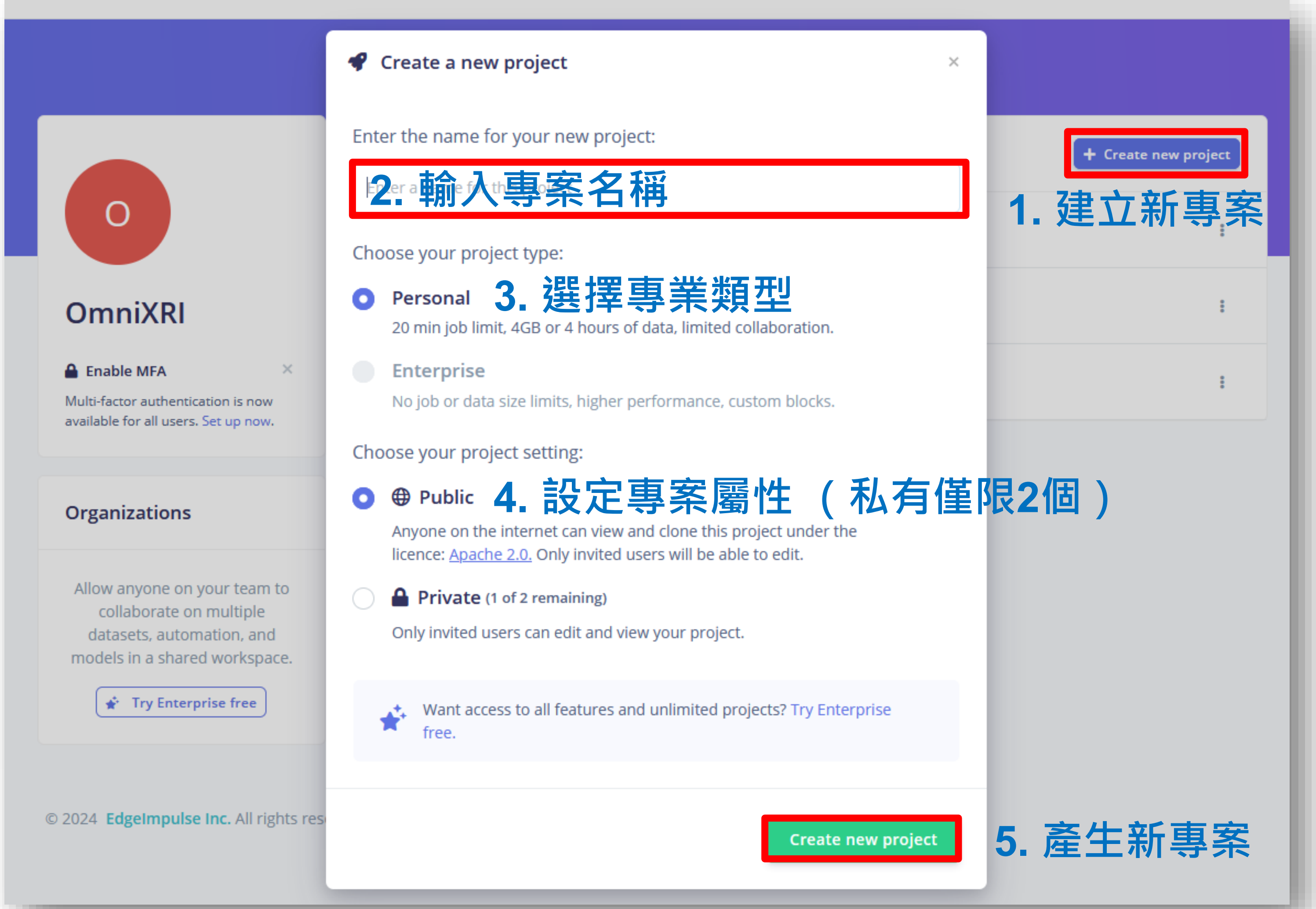
  npm install -g edge-impulse-cli --force

- 安裝完成後會得到下列工具
  - **edge-impulse-daemon 透過 COM 埠和雲端平台連接**
  - edge-impulse-uploader 允許上傳本機檔案
  - **edge-impulse-data-forwarder 透過 COM 埠上傳開發板感測器數值**
  - edge-impulse-run-impulse 顯示裝置上運行的模型
  - edge-impulse-blocks 建立自定義區塊
  - himax-flash-tool 燒錄 Haimax WE-I Plus

https://docs.edgeimpulse.com/docs/tools/edge-impulse-cli/cli-installation

建立 Edge Impulse 新專案





啟動edge-impulse-data-forwarder

C:\WINDOWS\system32\cmd.exe - "node" "C:\Users\jack_\AppData\Roaming\npm\\node_modules\edge-impulse-cli\build\cli\data-forward...

```
C:\Users\jack_>edge-impulse-data-forwarder --clean
Edge Impulse data forwarder v1.24.1
? What is your user name or e-mail address (edgeimpulse.com)? omnixri@gmail.com
? What is your password? [hidden]
Endpoints:
    Websocket: wss://remote-mgmt.edgeimpulse.com
    API:       https://studio.edgeimpulse.com
    Ingestion: https://ingestion.edgeimpulse.com

[SER] Connecting to COM8
[SER] Serial is connected (7F:63:29:74:0B:F4:EA:69)
[WS ] Connecting to wss://remote-mgmt.edgeimpulse.com
[WS ] Connected to wss://remote-mgmt.edgeimpulse.com

? To which project do you want to connect this device? OmniXRI / Xiao_nRF52840_IMU_Test
[SER] Detecting data frequency...
[SER] Detected data frequency: 130Hz
? 6 sensor axes detected (example values: [-0.02,-0.005,1.001,0.49,-1.4,0.84]). What do you want to call them? Separate
the names with ',': aX,aY,aX,gX,gY,gZ
[WS ] Device "XiaoSense" is now connected to project "Xiao_nRF52840_IMU_Test". To connect to another project, run `edge-
impulse-data-forwarder --clean`.
[WS ] Go to https://studio.edgeimpulse.com/studio/391754/acquisition/training to build your machine learning model!
[WS ] Incoming sampling request {
  path: '/api/training/data',
  label: 'OK',
  length: 10000,
  interval: 7.6923076923076925,
  hmacKey: 'd00f5918c1020b658b18569f9ee881c5',
  sensor: 'Sensor with 6 axes (aX, aY, aX, gX, gY, gZ)'
}
```

edge-impulse-data-forwarder --clean

輸入帳號

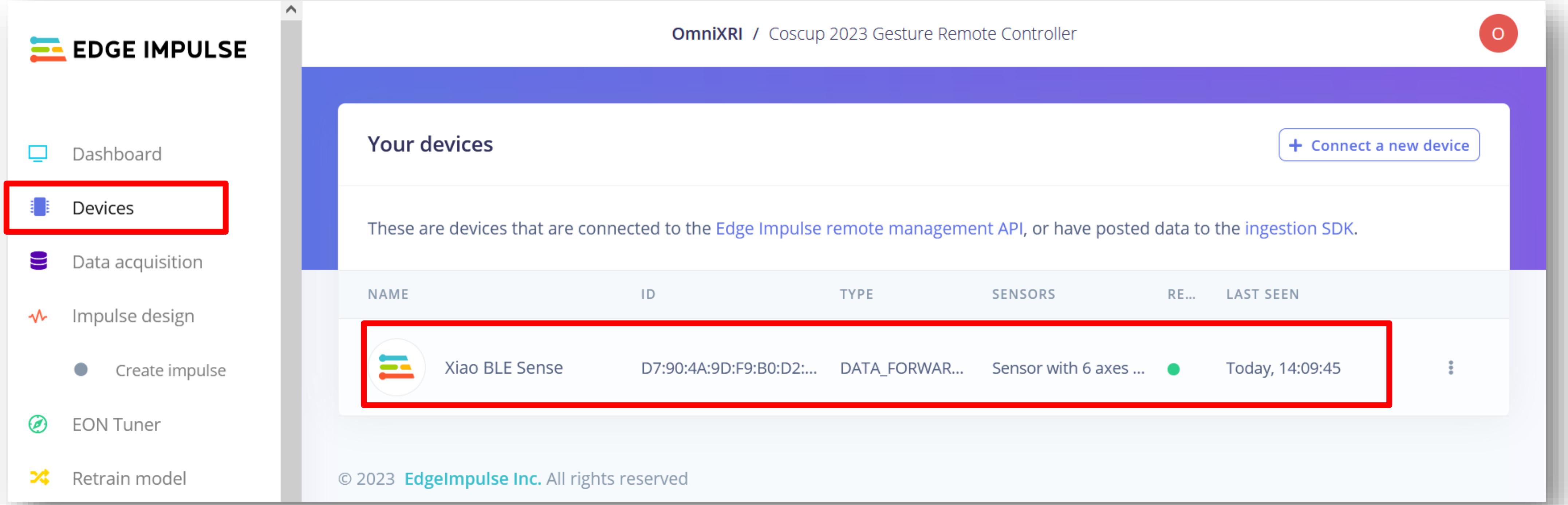
輸入密碼

選擇對應專案

輸入資料數量及名稱

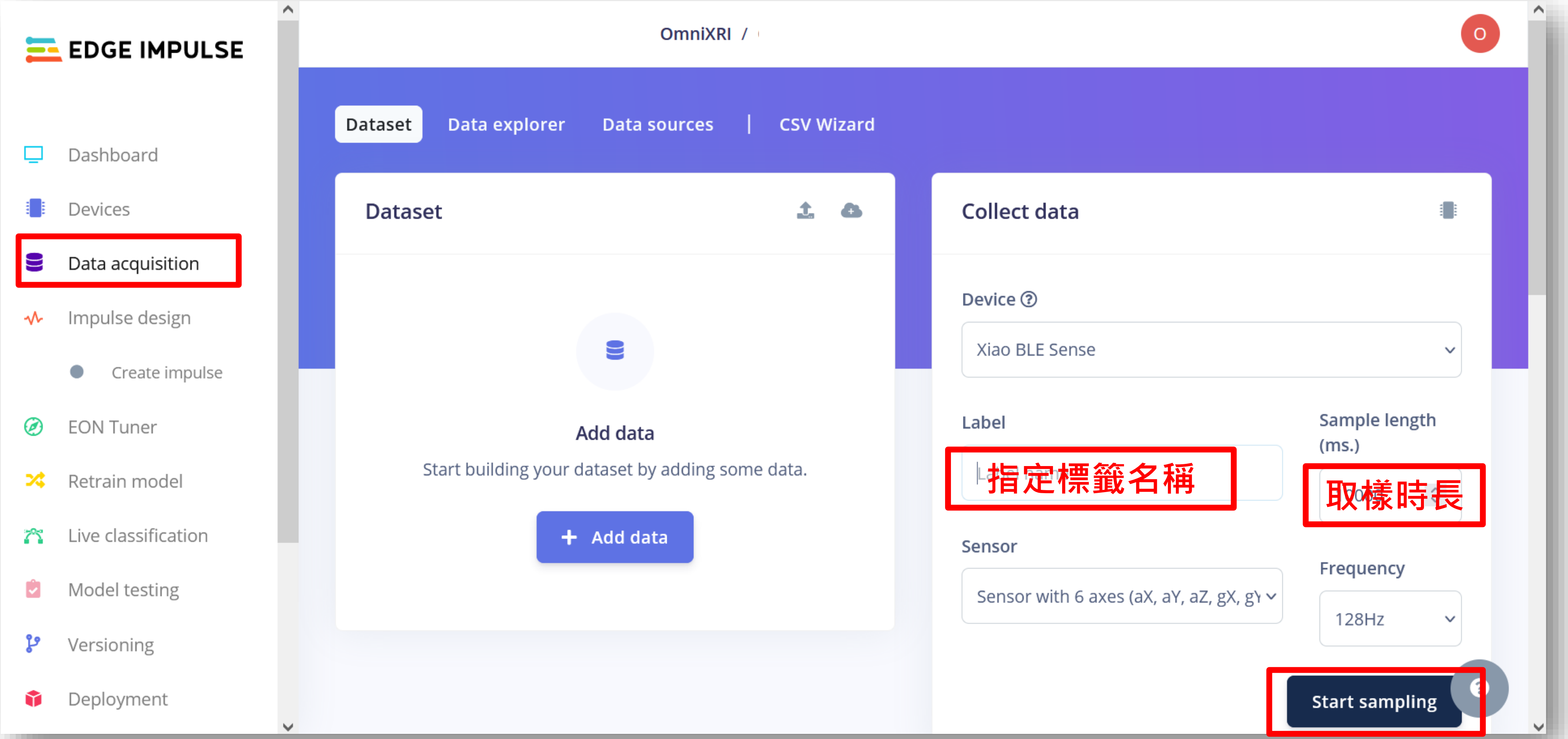
輸入帳號密碼並選定專案。準備接收來自開發板傳送的資料並給予並應的名稱。傳送的資料頻度必須大於2Hz。
若無法連接 COM 埠，請按開發板重置鍵一次，並請關閉 Arduino 及有可能佔用 COM 之程式。

檢查裝置是否已連線

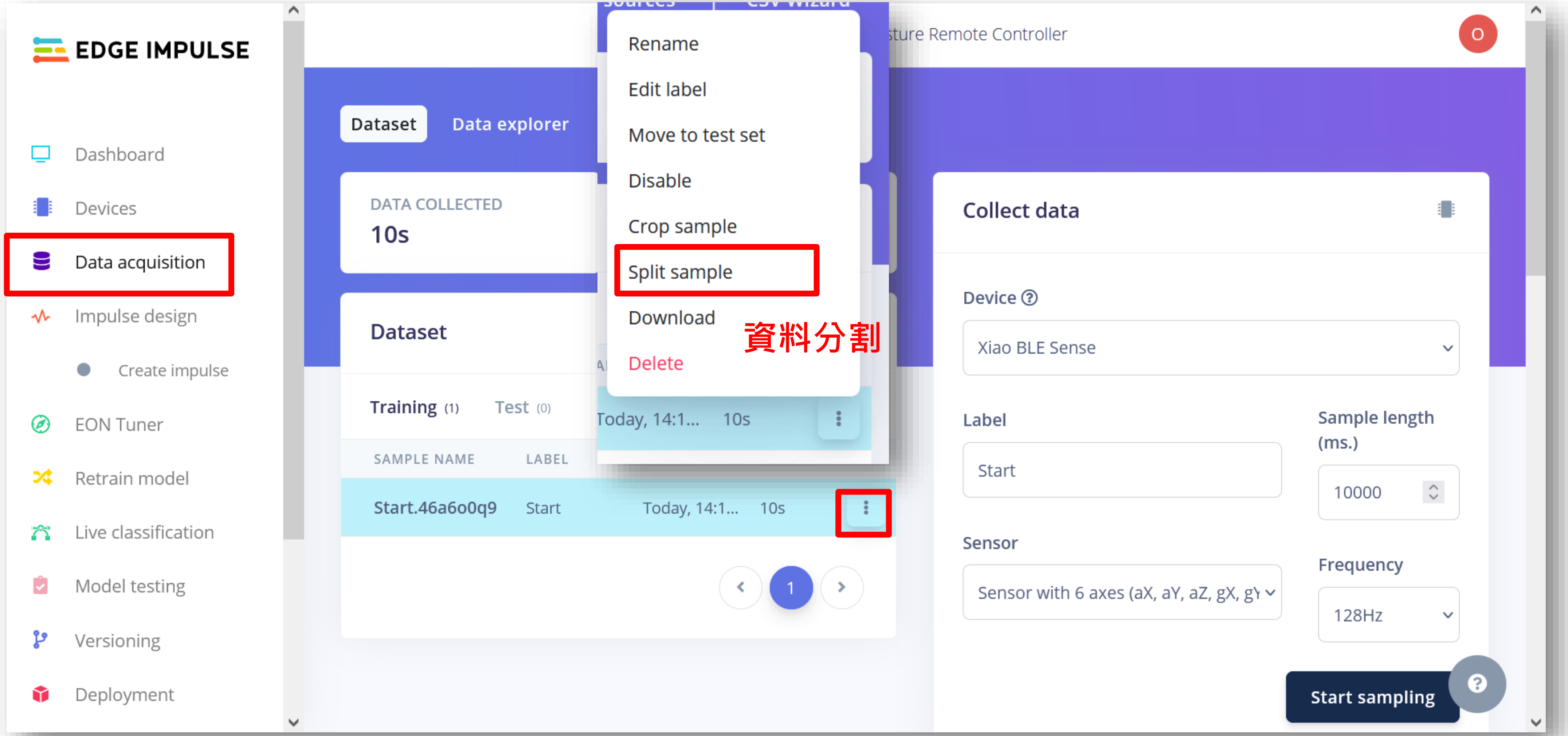


- 開發板運行IMU連續取值輸出程式。
- 啟動 edge-impulse-data-forwarder ，將裝置連線到雲端。
- 將開發板輸出之內容，傳送到雲端。

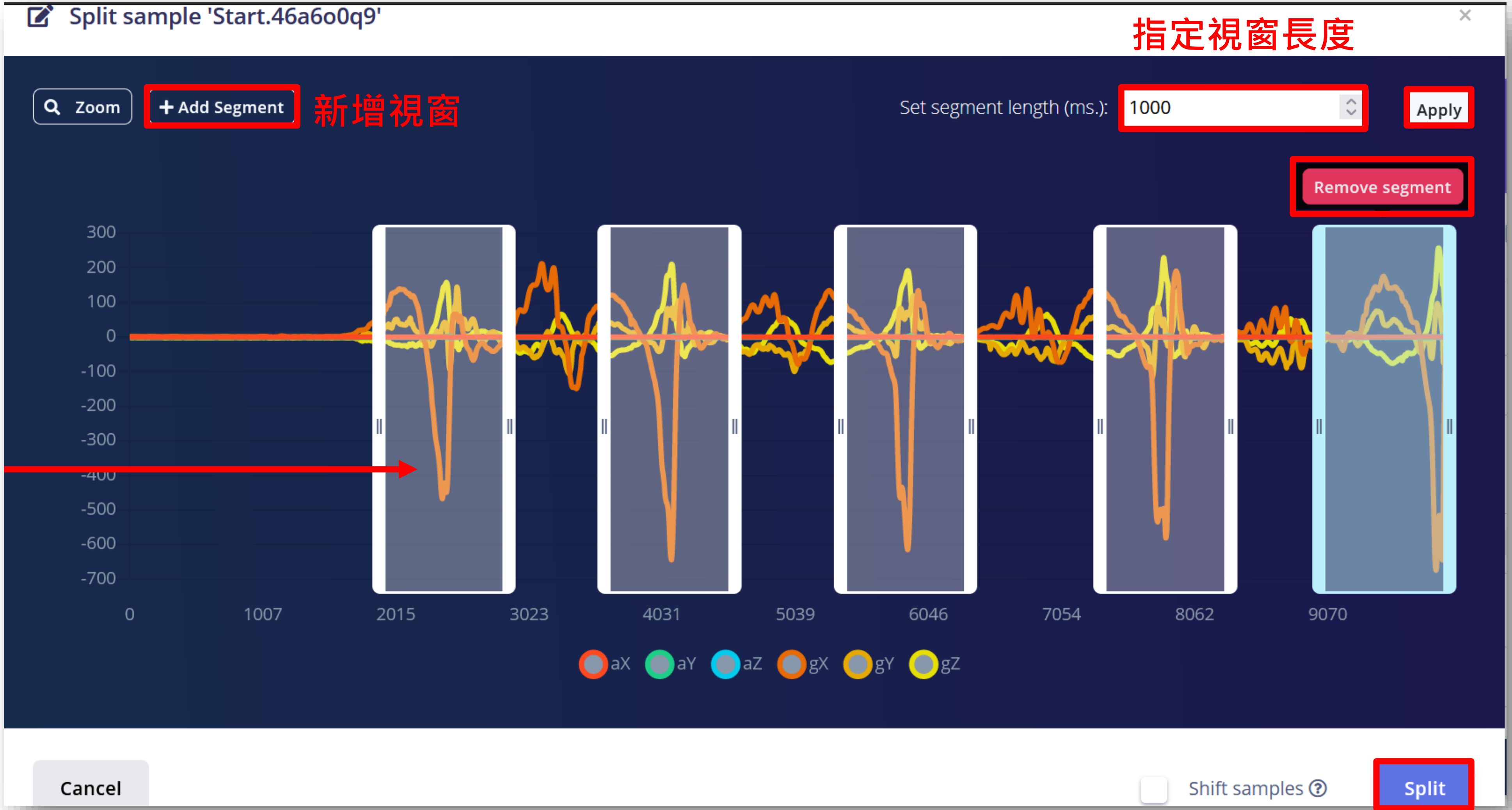
從外部裝置取得資料


EDGE IMPULSE
OmniXRI /
Dashboard
Devices
Data acquisition
Impulse design
Create impulse
EON Tuner
Retrain model
Live classification
Model testing
Versioning
Deployment
Dataset
Data explorer
Data sources
CSV Wizard
Dataset
Add data
Start building your dataset by adding some data.
+ Add data
Collect data
Device
Xiao BLE Sense
Label
指定標籤名稱
Sample length (ms.)
取樣時長
Sensor
Sensor with 6 axes (aX, aY, aZ, gX, gY
Frequency
128Hz
Start sampling

大量收集樣本並分割成獨立可訓練樣本



原始資料與自動分割



指定視窗長度

新增視窗

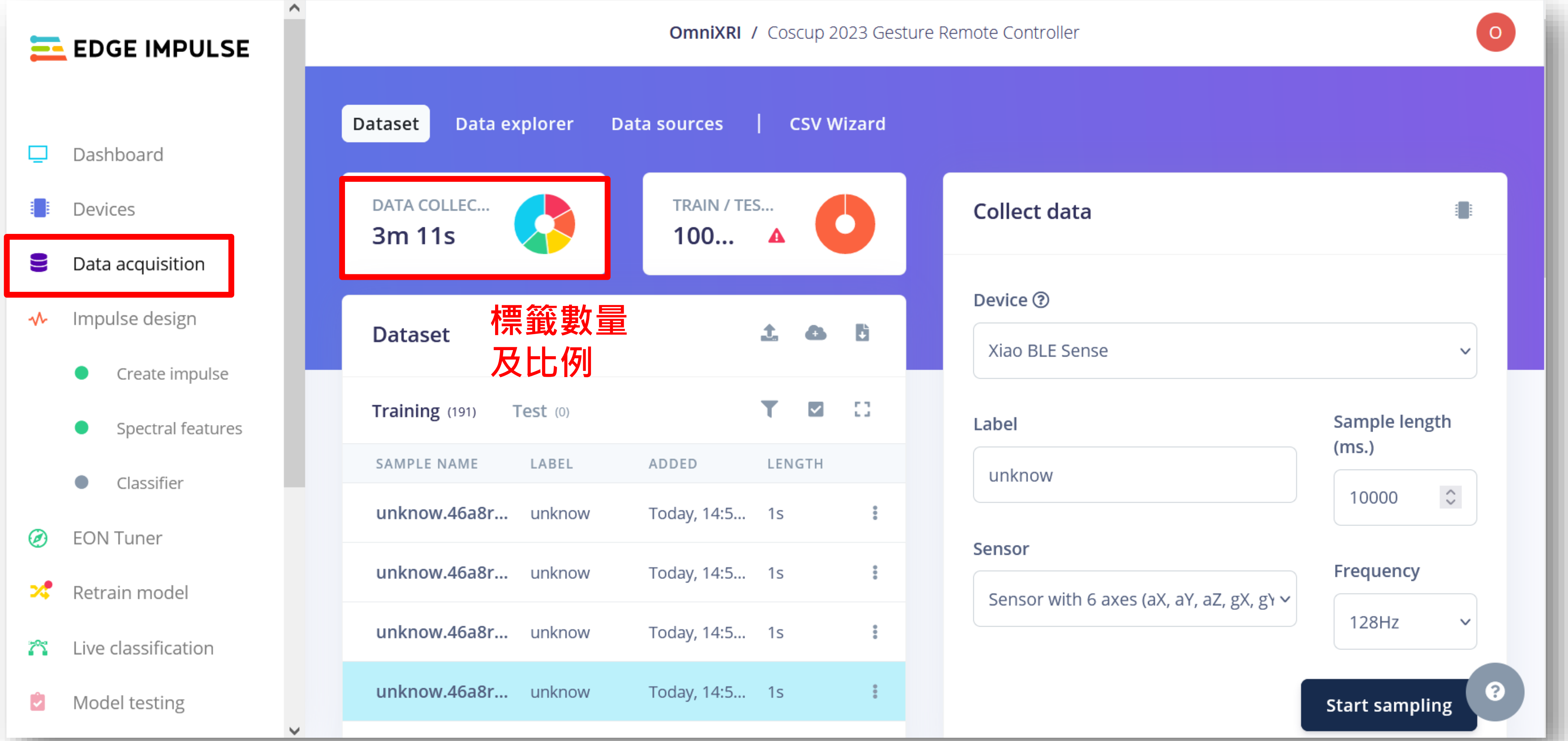
自動分割

刪除視窗

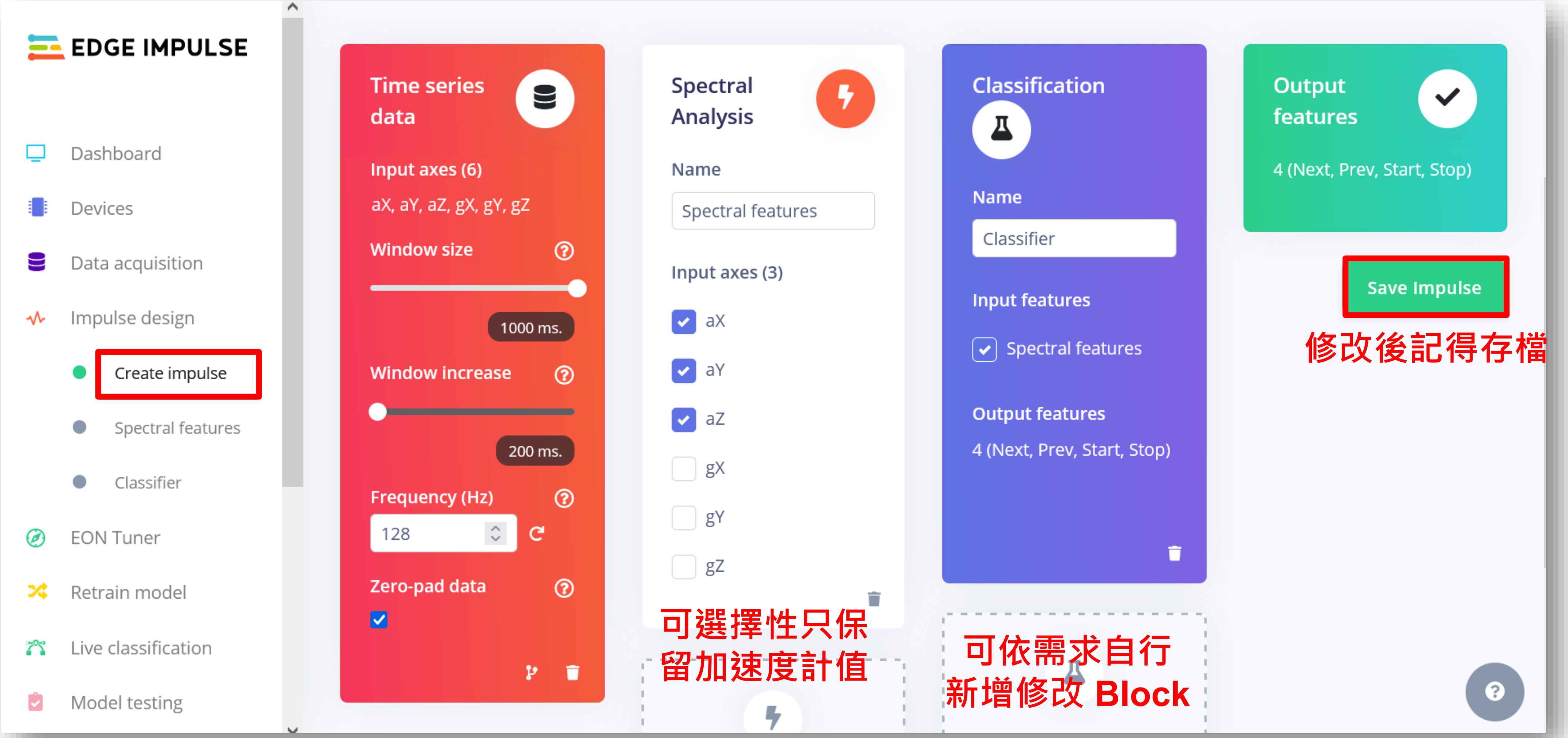
按住滑鼠左鍵不放可左右拖拉移動

完成分割

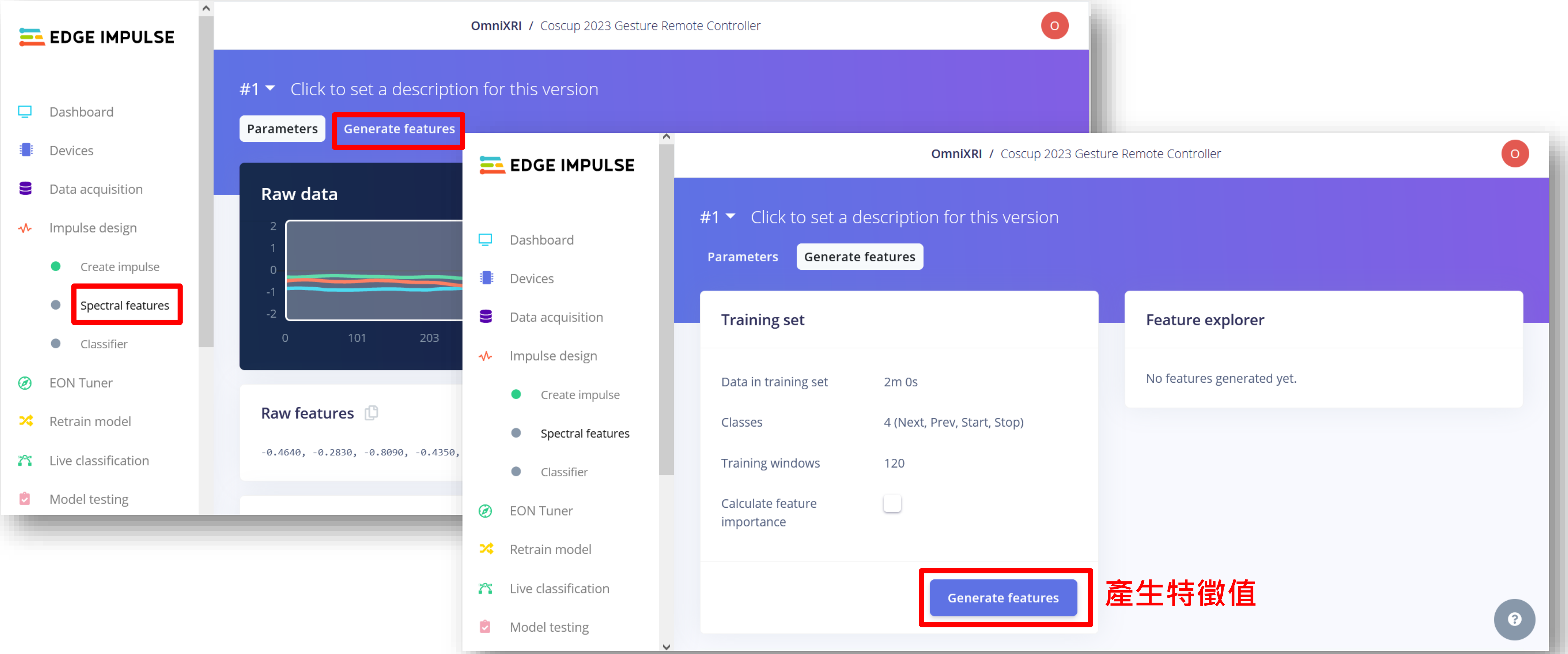
反復收集分割，建立完整資料集



選擇模型及設定必要參數



提取資料特徵

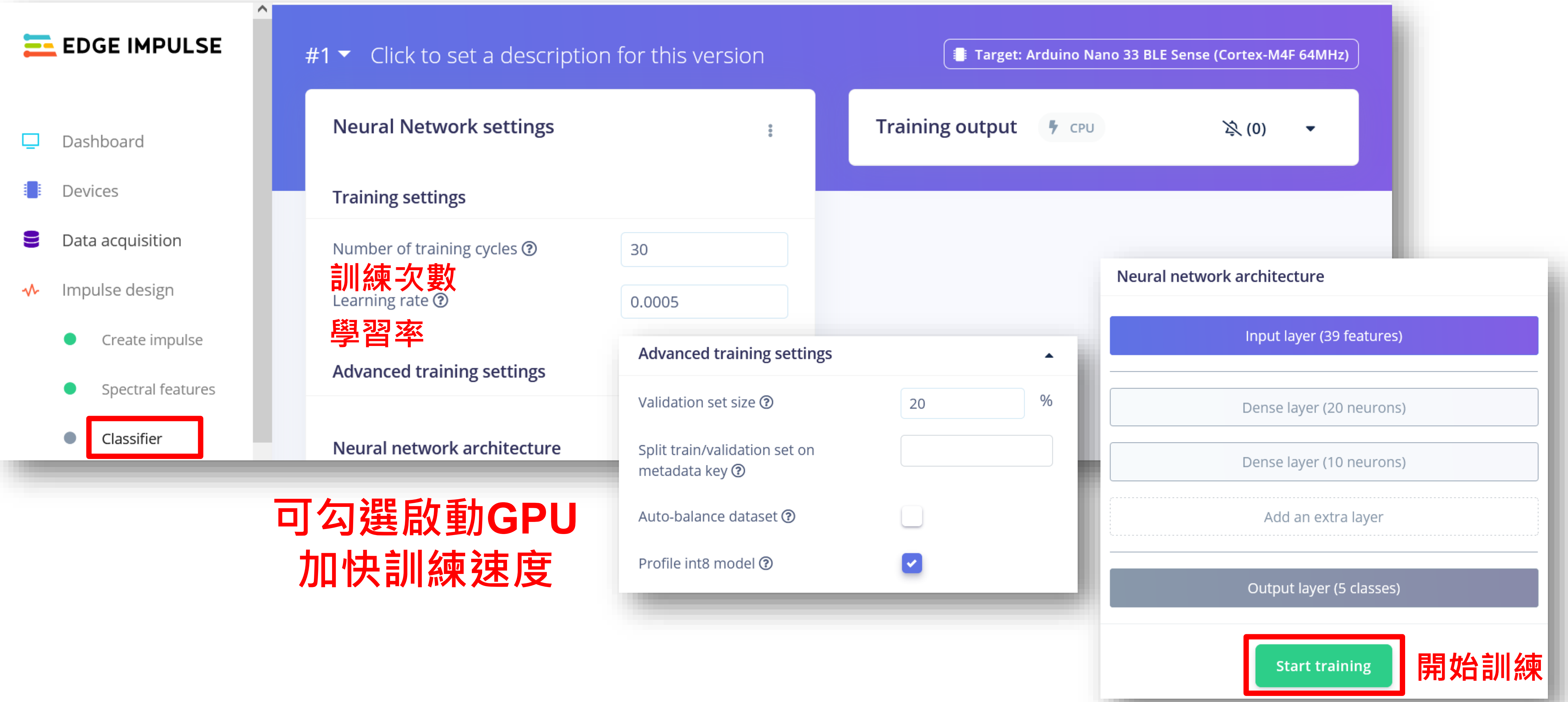


EDGE IMPULSE
Dashboard
Devices
Data acquisition
Impulse design
Create impulse
Spectral features
Classifier
EON Tuner
Retrain model
Live classification
Model testing
OmniXRI / Coscup 2023 Gesture Remote Controller
#1 Click to set a description for this version
Parameters
Generate features
Raw data
Raw features
-0.4640, -0.2830, -0.8090, -0.4350,
Training set
Data in training set 2m 0s
Classes 4 (Next, Prev, Start, Stop)
Training windows 120
Calculate feature importance
Feature explorer
No features generated yet.
Generate features
產生特徵值

提取特徵結果

以指定的硬體進行資源
使用及推論時間估測

設定分類訓練相關參數



可勾選啟動GPU
加快訓練速度

開始進行模型訓練及結果顯示

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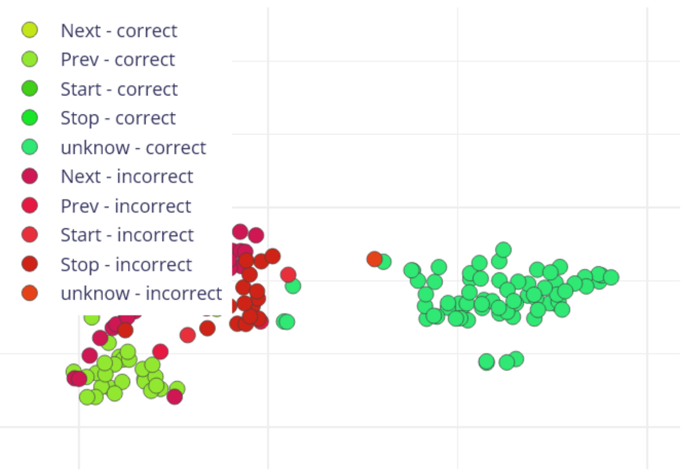
NATIONAL TAIWAN UNIVERSITY OF SCIENCE AND TECHNOLOGY

Training output CPU (0)

```
INFO: Created TensorFlow Lite XNNPACK delegate for CPU.
Calculating inferencing time OK
Profiling float32 model...
Profiling float32 model (TensorFlow Lite Micro)...
Profiling float32 model (EON)...
Attached to job 11100469...
Attached to job 11100469...
Profiling int8 model...
Profiling int8 model (TensorFlow Lite Micro)...
Profiling int8 model (EON)...
Attached to job 11100469...
Attached to job 11100469...

Model training complete

Job completed
```

Data explorer (full training set)



On-device performance

INFERE...	PEAK RA...	FLASH U...
1 ms.	1.3K	14.7K

Model Model version: Quantized (int8)

Last training performance (validation set)

ACCURACY 82.1% LOSS 0.80

Confusion matrix (validation set)

	NEXT	PREV	START	STOP	UNKNOW
NEXT	33.3%	50%	0%	16.7%	0%
PREV	0%	100%	0%	0%	0%
START	0%	0%	100%	0%	0%
STOP	20%	20%	0%	40%	20%
UNKNOW	0%	0%	0%	0%	100%
F1 SCORE	0.44	0.75	1.00	0.50	0.97

30 Epoch 結果

Last training performance (validation set)

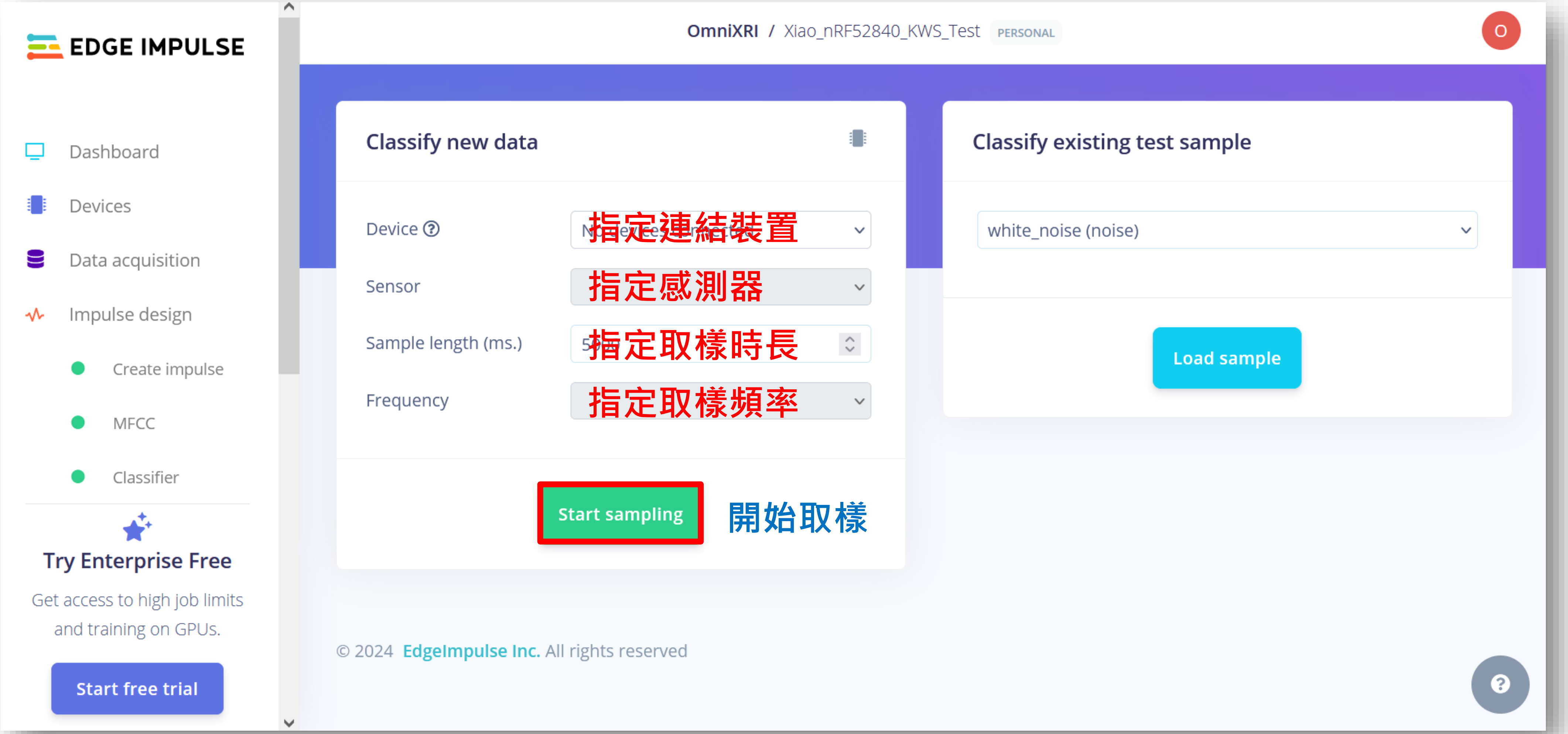
ACCURACY 94.9% LOSS 0.26

Confusion matrix (validation set)

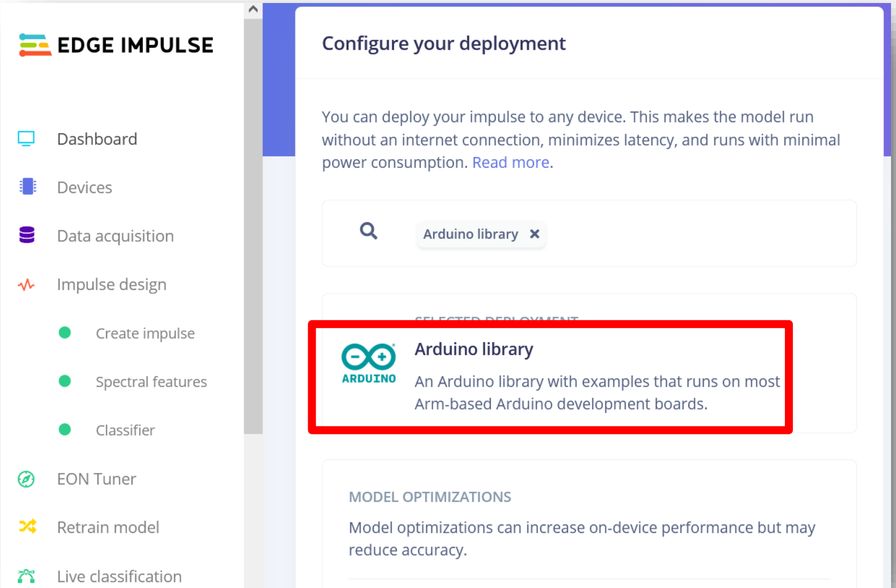
	NEXT	PREV	START	STOP	UNKNOW
NEXT	83.3%	16.7%	0%	0%	0%
PREV	0%	100%	0%	0%	0%
START	0%	0%	100%	0%	0%
STOP	20%	0%	0%	80%	0%
UNKNOW	0%	0%	0%	0%	100%
F1 SCORE	0.83	0.92	1.00	0.89	1.00

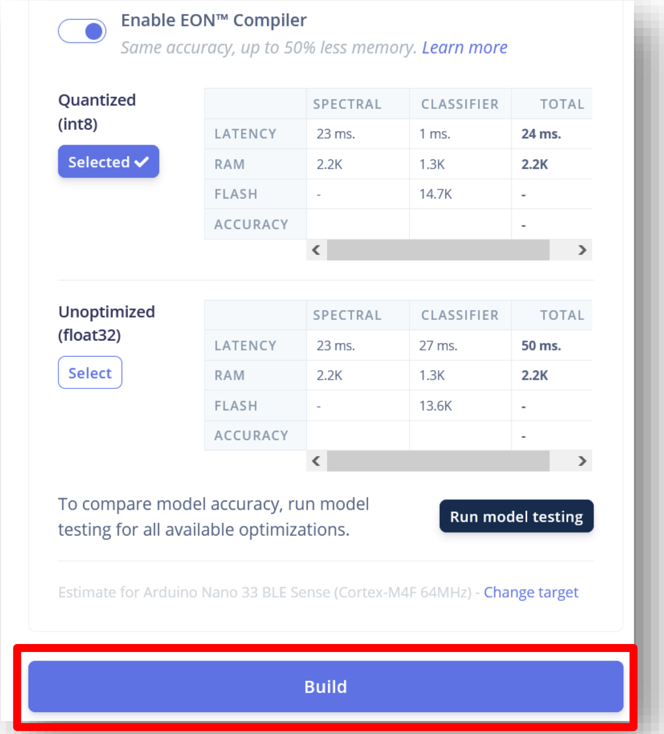
100 Epoch 結果

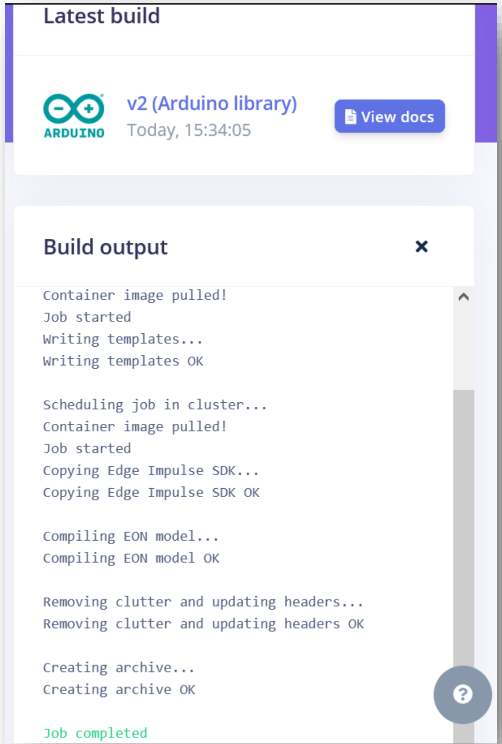
線上測試（從外部裝置取樣）

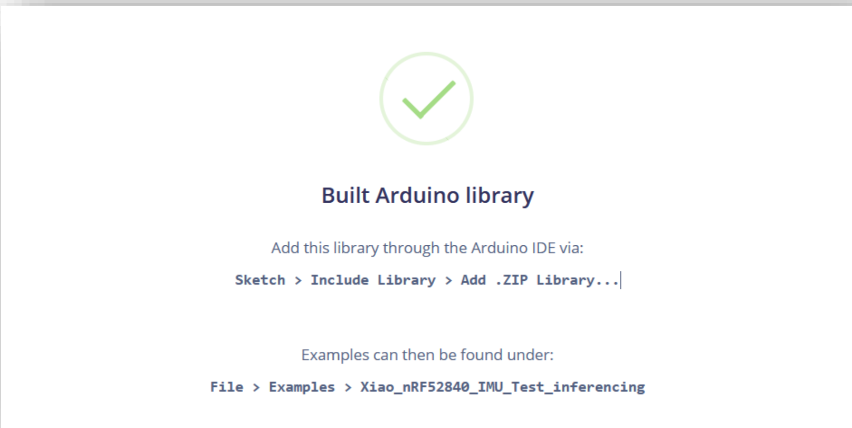


選擇部署種類及設定參數









以 Arduino 函式庫(*.zip)方式輸出
預設檔名為「ei-專案名稱-arduino-版本序號.zip」

導入 Arduino 函式庫並進行推論測試(1/2)

- 新增函式庫 **Sketch > Include Library > Add .ZIP Library …**
- 新增範例 **File > Examples > ei-專案名稱_inferencing > nano_ble33_sense > nano_ble33_sense_accelerometer（原始範例輸出結果從 COM 輸出文字串）**
- 經編譯會在「**#error "Invalid model for current sensor"**」這列出現「**"invalid model for current sensor"**」，主要原因為Edge Impulse打包時是以Sensor_Fusion方式處理而不是Accelerometer方式造成。
- 點擊選單 **File > Preferences** 找到 Sketchbook Folder，進到該路徑後，進到 \libraries\專案名稱_inferencing\src\model-parameters\ 開啟 **model_metadata.h** 。找到「 #define EI_CLASSIFIER_SENSOR EI_CLASSIFIER_SENSOR_**FUSION** 」，將定義改成「 EI_CLASSIFIER_SENSOR_**ACCELEROMETER** 」存檔即可。

導入 Arduino 函式庫並進行推論測試(2/2)

- 回到 Arduino 範例後會發現使用的運動感測器不同，所以需將「**#include <Arduino_LSM9DS1.h>**」註解掉並新增下列程式

  #include <LSM6DS3.h>

  #include <Wire.h>

  LSM6DS3 myIMU(I2C_MODE, 0x6A);

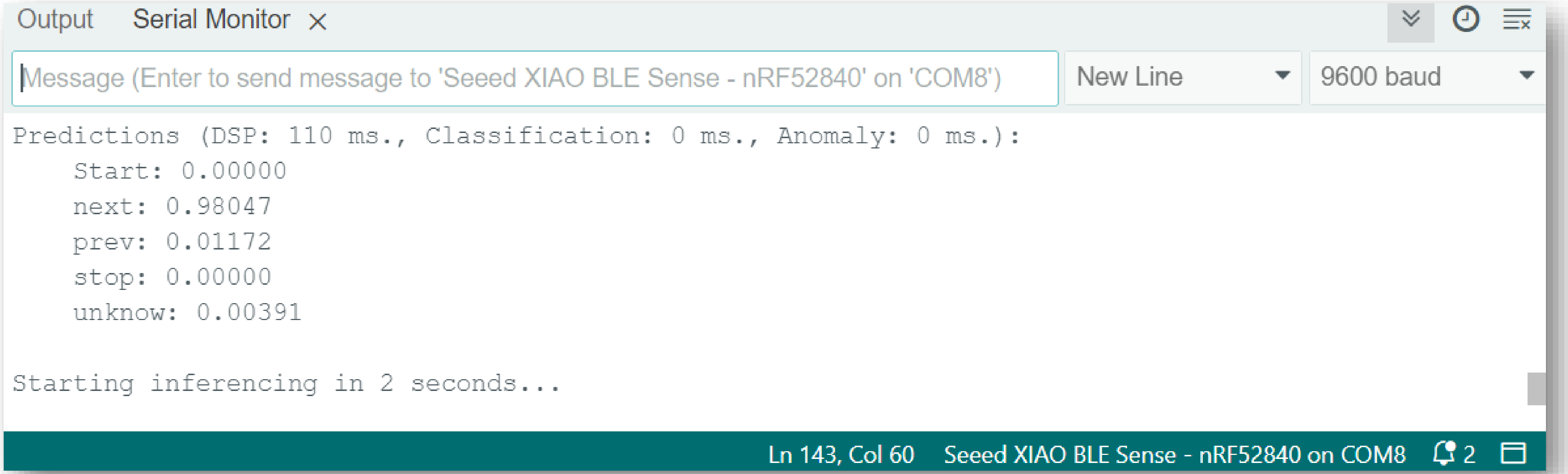
```
/* Includes ---------------------------------------------------------------
#include <Xiao_nRF52840_IMU_Test_inferencing.h>
//#include <Arduino_LSM9DS1.h> //Click here to get the libr
#include <LSM6DS3.h>
#include <Wire.h>
/* Constant defines -------------------------------------------------------
#define CONVERT_G_TO_MS2    9.80665f
#define MAX_ACCEPTED_RANGE  2.0f        // starting 03/2022

LSM6DS3 myIMU(I2C_MODE, 0x6A);    //I2C device address 0x6A
```

- 在Setup函式中的初始化也要將「if (!IMU.begin()) {」修改為「 if (!**myIMU**.begin()) { 」
- 另外由於讀取IMU運動感測器函式不同因此也需修改如左。

```
//        IMU.readAcceleration(buffer[ix], buffer[ix + 1],
        buffer[ix]     = myIMU.readFloatAccelX();
        buffer[ix + 1] = myIMU.readFloatAccelY();
        buffer[ix + 2] = myIMU.readFloatAccelZ();
```

- 重新編譯前確認開發板是設定在「 **Seeed nRF52 mbed-enabled Boards** > Seeed XIAO BLE Sense - nRF52840 」，不然程式運行後會找不到IMU LSM6DS3。

手勢辨識結果（Arduino部份）



- 推論結果，包括前處理(DSP)、分類及異常時間 (ms)
- 將五個標籤的置信度顯示出來，1.0為完全正確，0.0為完全不正確，可設置門檻值來限制置信度不高之答案。
- **No Code型式**，範例程式直接編譯上傳，不需改寫任何程式，輸出結果由 COM 接收文字串。

Arduino 2.0 快速(增量)編譯

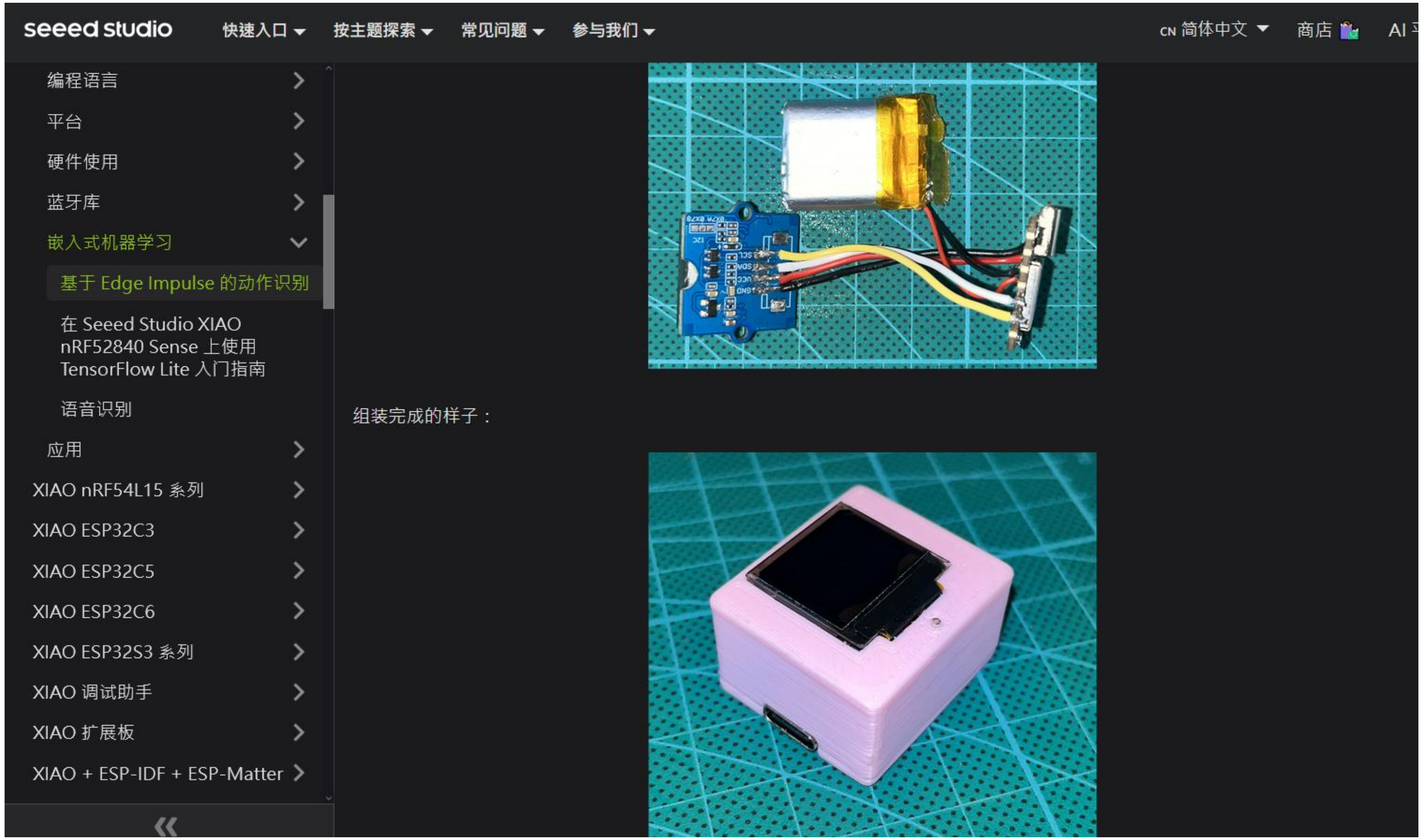
非常重要：

強烈建議一定要採用Arduino CLI 增量編譯方式更新程式修改，否則每次將耗費大量時間，增加碳排量。

工作內容	Arduino IDE	Arduino CLI
第一次編譯上傳	19~20min	19~20min
未修改程式第二次編譯上傳	19~20min	< 1min
修改程式後編譯上傳	19~20min	1~2min

https://omnixri.blogspot.com/2024/10/arduino-20.html

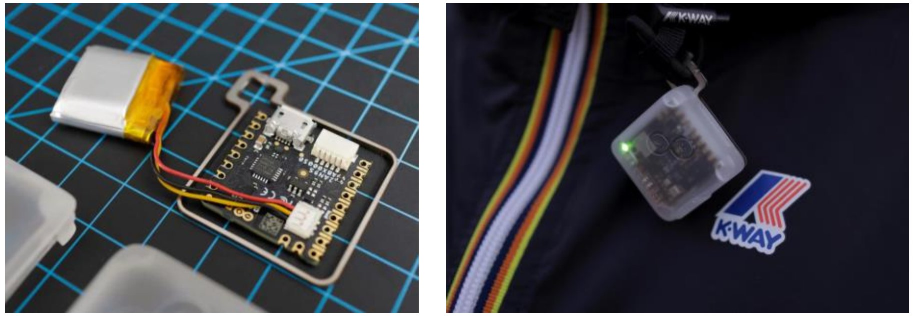
nRF52840 基於Edge Impulse 的動作辨識



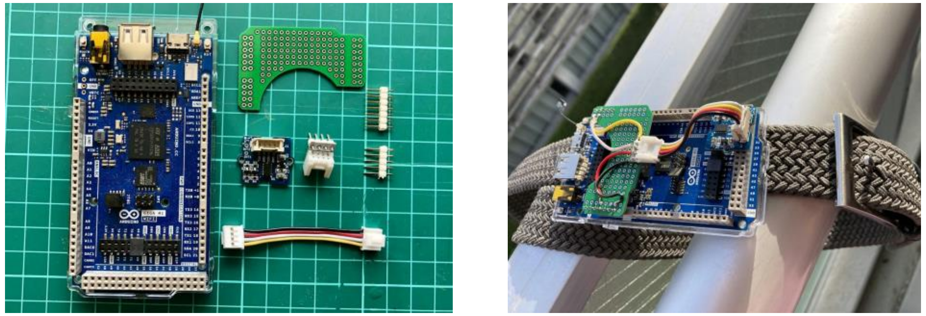
https://wiki.seeedstudio.com/cn/XIAOEI/

Edge Impulse Expert 案例分享 (1/2)

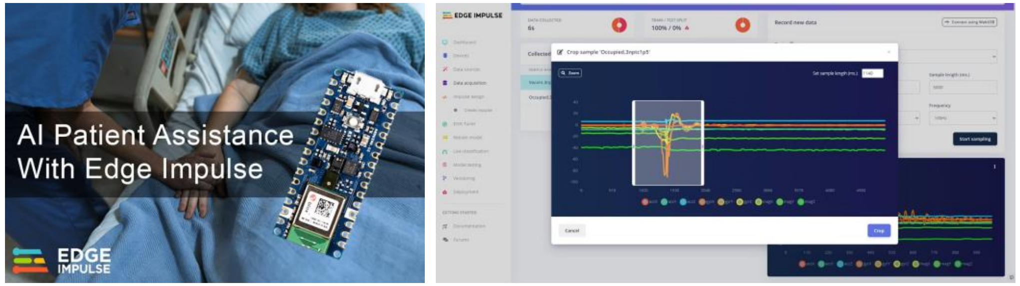
Arduino x K-Way - Outdoor Activity Tracker



Fall Detection using a Transformer Model – Arduino Giga R1 WiFi



Hospital Bed Occupancy Detection - Arduino Nano 33 BLE Sense

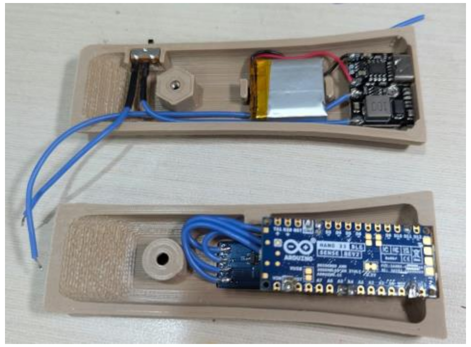


Bluetooth Fall Detection - Arduino Nano 33 BLE Sense

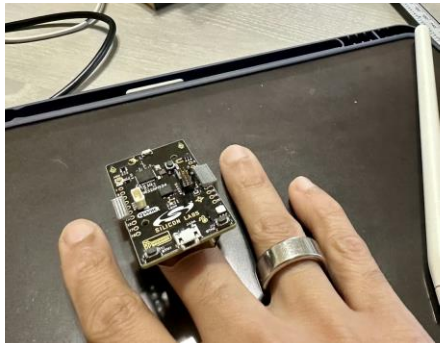
https://docs.edgeimpulse.com/projects/expert-network/project-list#accelerometer-and-activity-projects

Edge Impulse Expert 案例分享 (2/2)

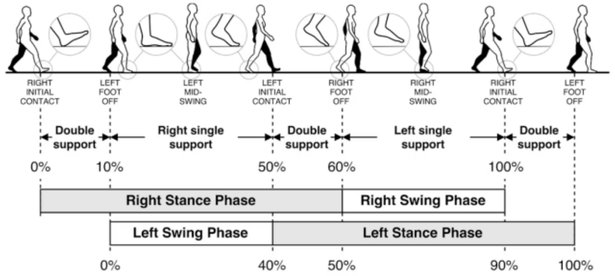
Smart Table Tennis Bat with Live Analytics - Arduino Nano 33 BLE Sense



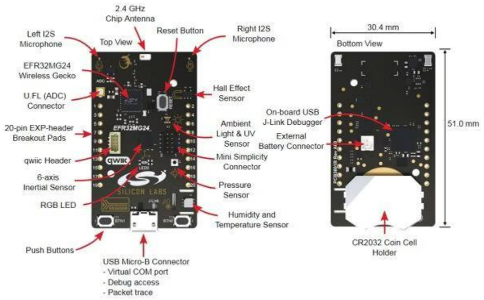
Porting a Gesture Recognition Project from the SiLabs Thunderboard Sense 2 to xG24

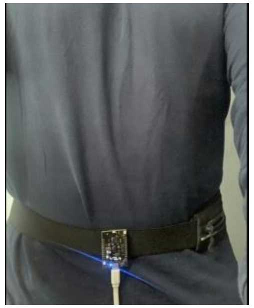


Continuous Gait Monitor (Anomaly Detection) - Nordic Thingy:53



Porting a Posture Detection Project from the SiLabs Thunderboard Sense 2 to xG24

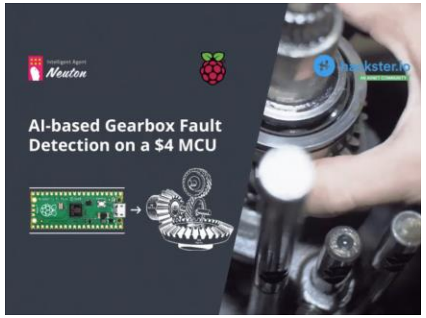




https://docs.edgeimpulse.com/projects/expert-network/project-list#accelerometer-and-activity-projects

其它案例分享(1/2)

齒輪箱異常振動偵測



【功能簡述】
利用四個振動感測器產生的資料來分析變速箱是否健康，透過現成資料集進行模型訓練，完成後再部署到開發板上測試，未接真的感測器。

【硬體/週邊】 Raspberry Pi Pico
【軟體/平台】 Neuton Tiny ML Neuton + Arduino IDE

https://www.hackster.io/mech/tinyml-gearbox-fault-prediction-on-a-4-mcu-41f434

包裹運送異常震動偵測

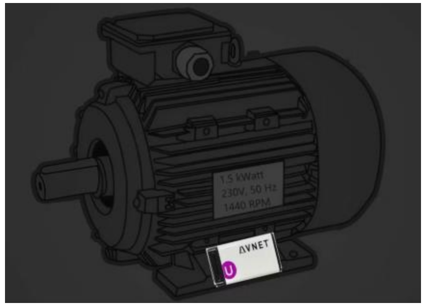
【功能簡述】
利用運動感測器來偵測包裹運送途中的異常狀態，包括亂摔、掉落、搖晃等。

【硬體/週邊】 Arduino Nano 33 BLE Sense + SIM 800L GSM Module
【軟體/平台】 Edge Impulse Studio + Arduino IDE

https://blog.arduino.cc/2022/09/10/this-tinyml-device-monitors-packages-for-damage-while-in-transit/

其它案例分享(2/2)

馬達異常振動偵測



【功能簡述】
利用開發板上運動感測器產生的資料來分析馬達是否健康，包括

- 軸承故障（例如刺耳的啟動噪音和更大的運轉噪音）
- 對準問題（例如金屬刮擦噪音）
- 機械負載損失（例如不抽水）
- 冷卻風扇損壞（例如風扇葉片在金屬蓋上刮傷）

【硬體/週邊】 Avnet SmartEdge Agile Brainium
【軟體/平台】 Avnet Brainium

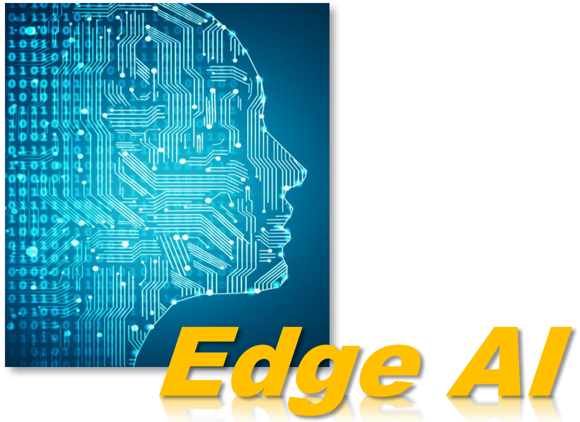
https://www.hackster.io/PSoC_Rocks/industrial-motor-diagnostic-866e96

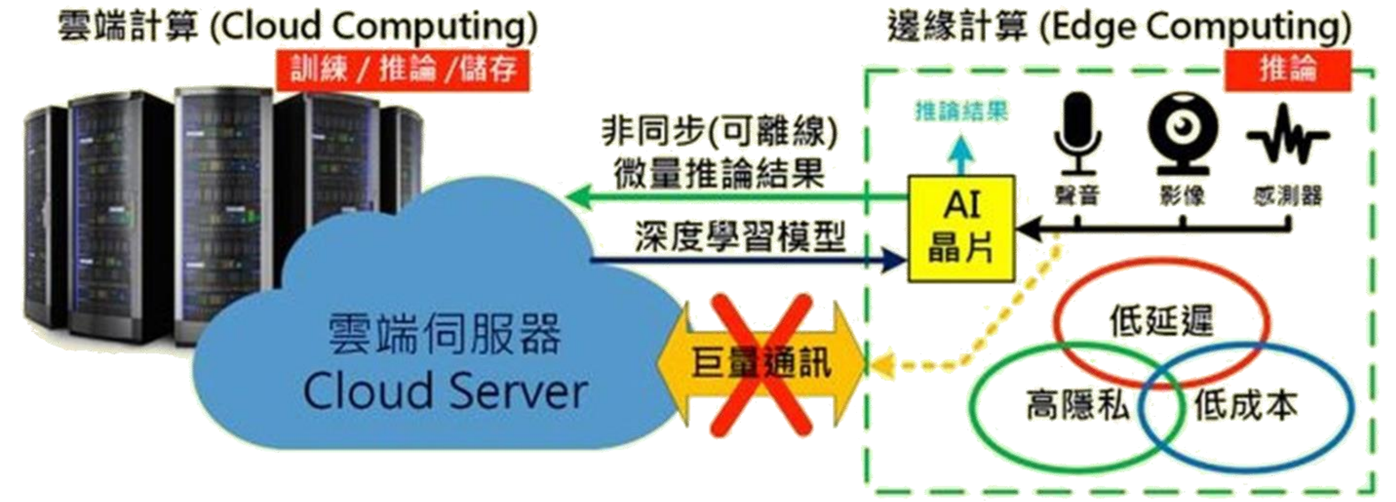
智能桌球拍（虛擬教練）

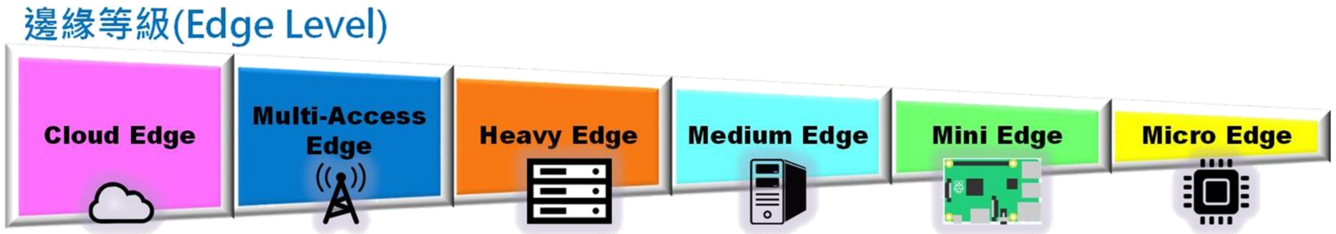
【功能簡述】
在桌球拍上安裝運動感測器來記錄及分析揮拍姿勢，以利改善後續表現。

【硬體/週邊】 Arduino Nano 33 BLE Sense + 桌球拍
【軟體/平台】TensorFlow Lite for Microcontrollers + Google Creativelab - Tiny Motion Trainer + Arduino IDE

https://blog.arduino.cc/2022/09/10/this-tinyml-device-monitors-packages-for-damage-while-in-transit/


Edge AI


雲端計算 (Cloud Computing)
訓練 / 推論 /儲存
邊緣計算 (Edge Computing)
推論
推論結果
非同步(可離線)
微量推論結果
深度學習模型
AI
晶片
聲音
影像
感測器
雲端伺服器
Cloud Server
巨量通訊
低延遲
高隱私
低成本


邊緣等級(Edge Level)
Cloud Edge
Multi-Access Edge
Heavy Edge
Medium Edge
Mini Edge
Micro Edge

9.3. 異常振動偵測

時間序列信號預測

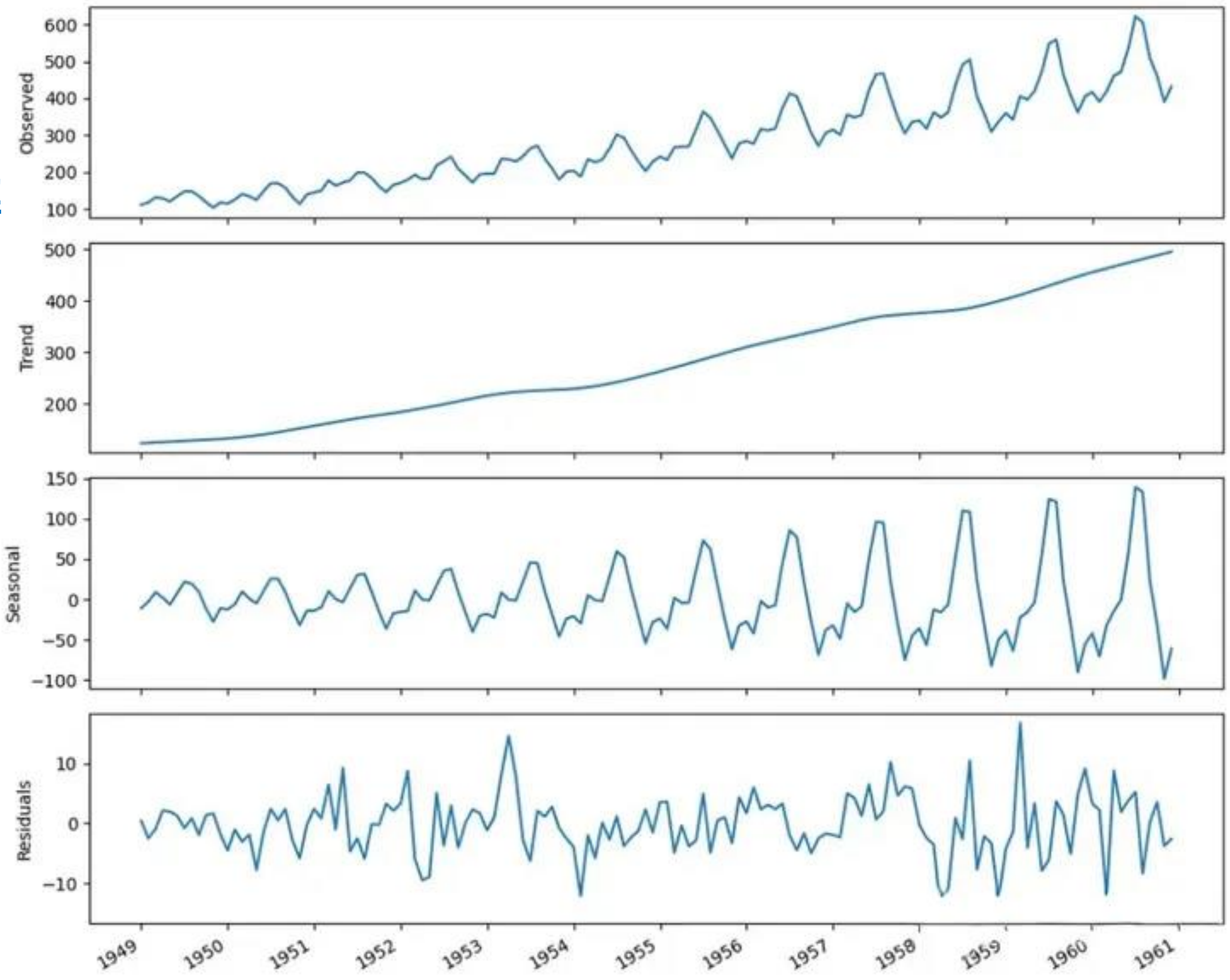
Time Series
原始信號

分解

趨勢

季節性

殘差

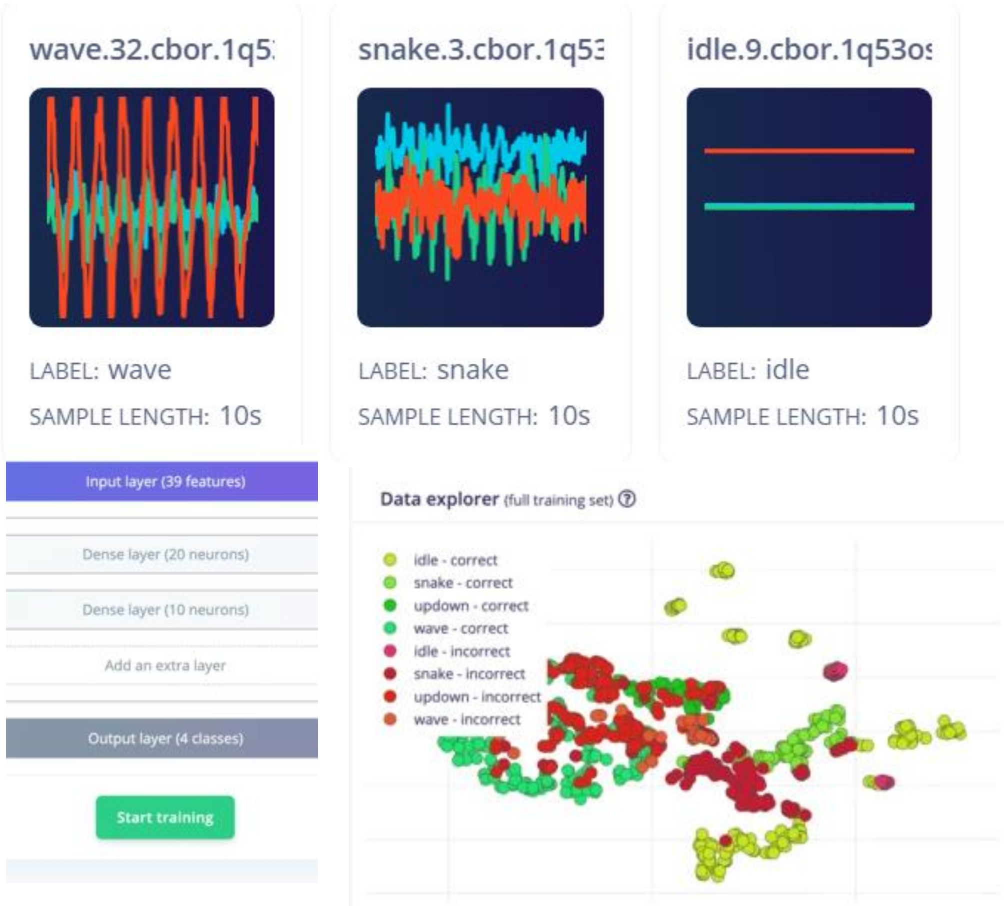


資料來源：https://vocus.cc/article/6927b46cfd897800014271ee

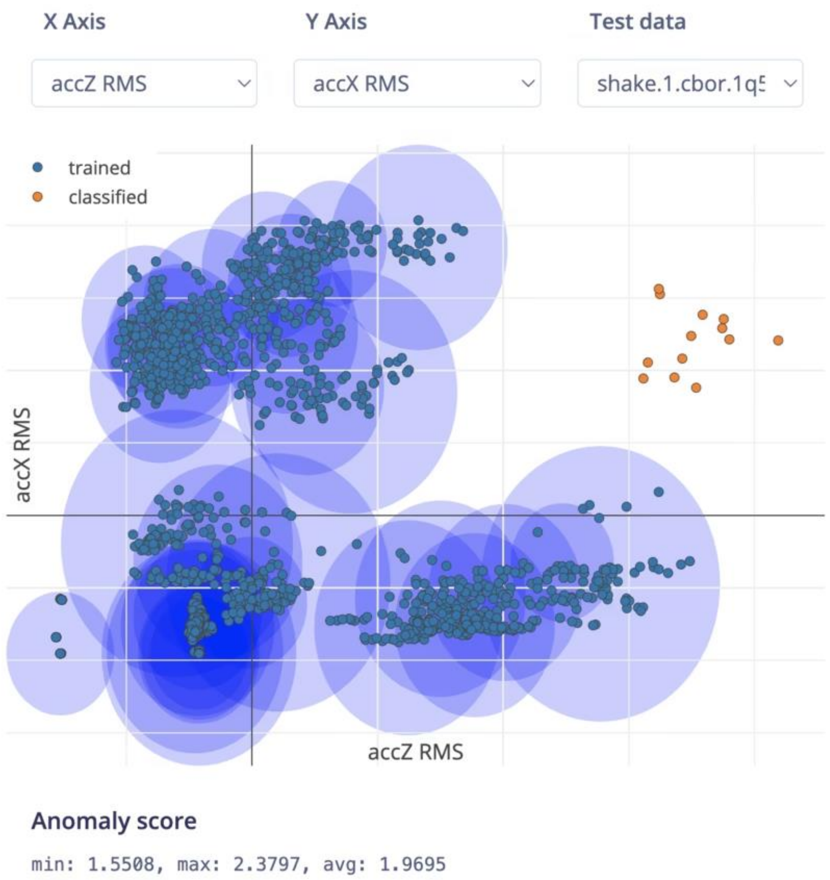
- **傳統時間序列預測模型**
  - 自迴歸(AR)
  - 移動平均(MA)
  - 自迴歸移動平均(ARMA)
  - 自迴歸綜合移動平均(ARIMA)
  - 季節自迴歸綜合移動平均(SARIMA)
  - 帶有外源迴歸量的季節自迴歸綜合移動平均(SARIMAX)
  - 向量自迴歸(VAR)
  - 向量誤差校正(VECM)
- **深度學習時間序列預測模型**
  - 多層感知器(MLP)
  - 迴圈神經網路(RNN)
  - 長短期記憶網路(LSTM)
  - 自迴歸 LSTMs

常見時序信號異常偵測方式

- 特徵樣版分類

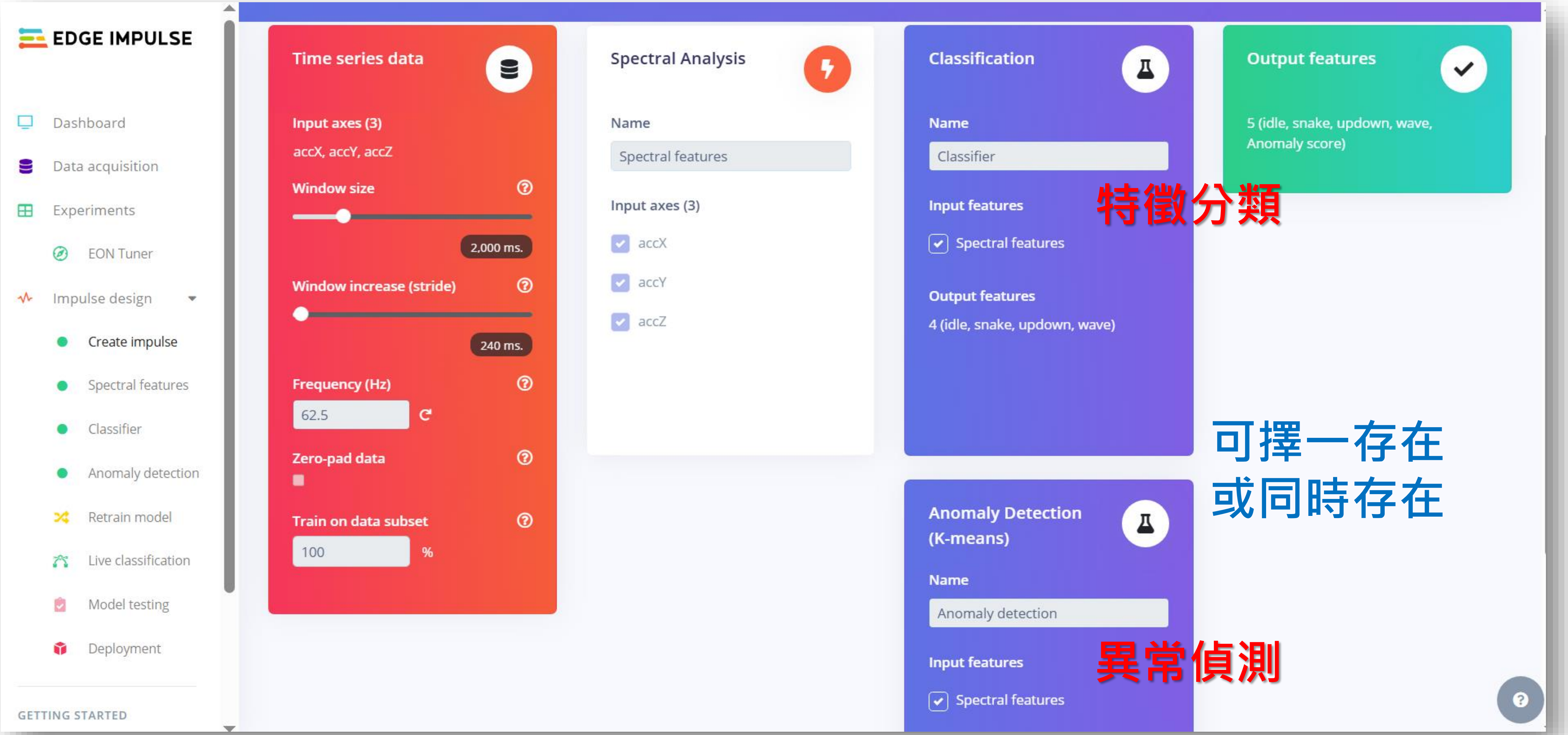


- 聚類離群偵測



資料來源：https://docs.edgeimpulse.com/tutorials/end-to-end/motion-recognition

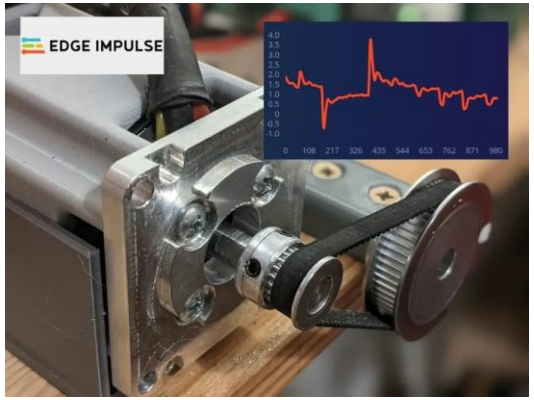
Edge Impulse 專案 – Impulse Design

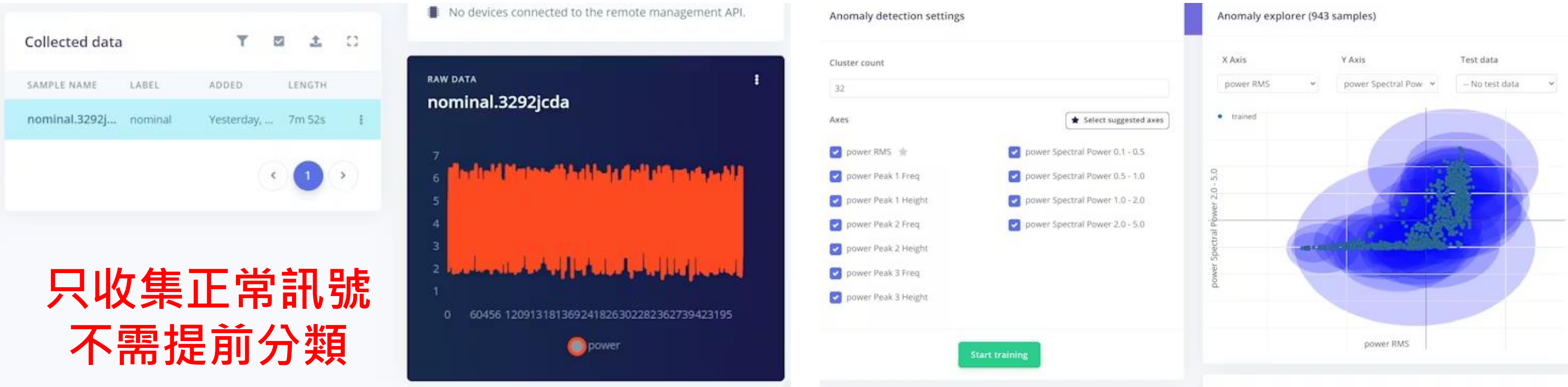


可擇一存在
或同時存在

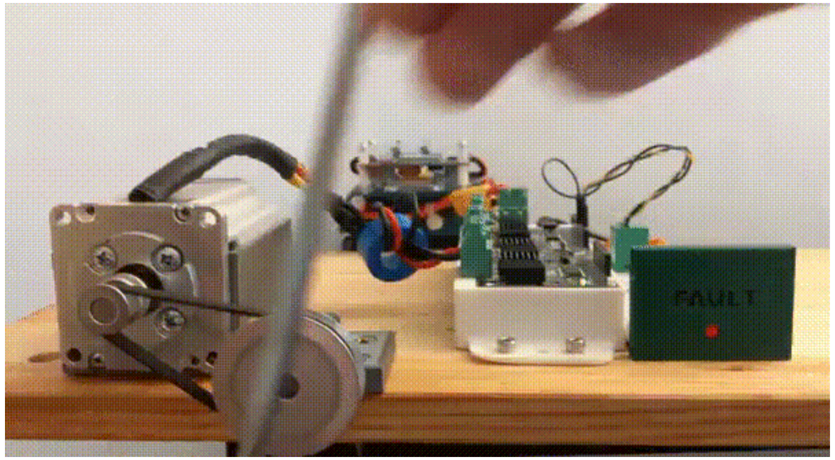
https://studio.edgeimpulse.com/public/14299/latest/impulse/1/create-impulse

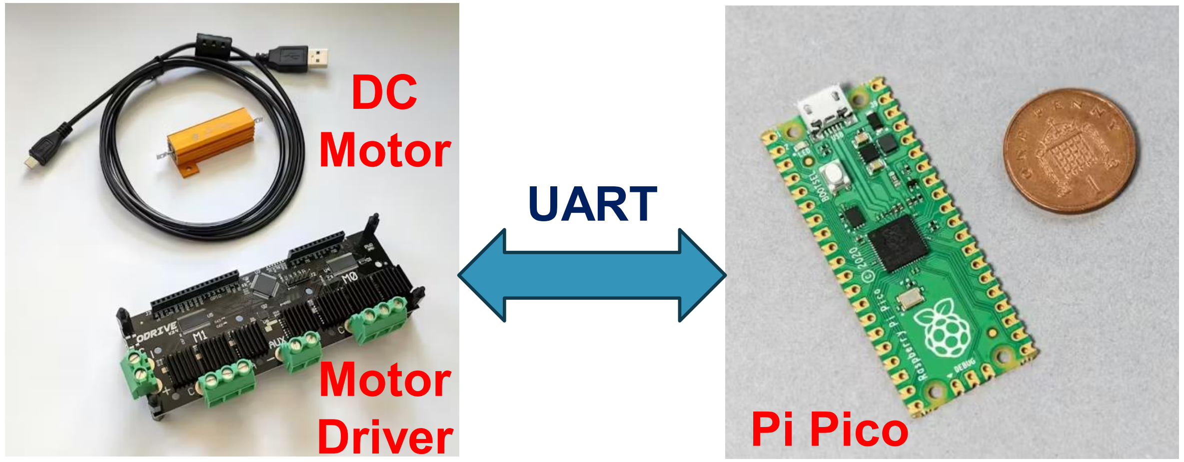
練習案例 – 無刷直流馬達異常偵測





只收集正常訊號
不需提前分類

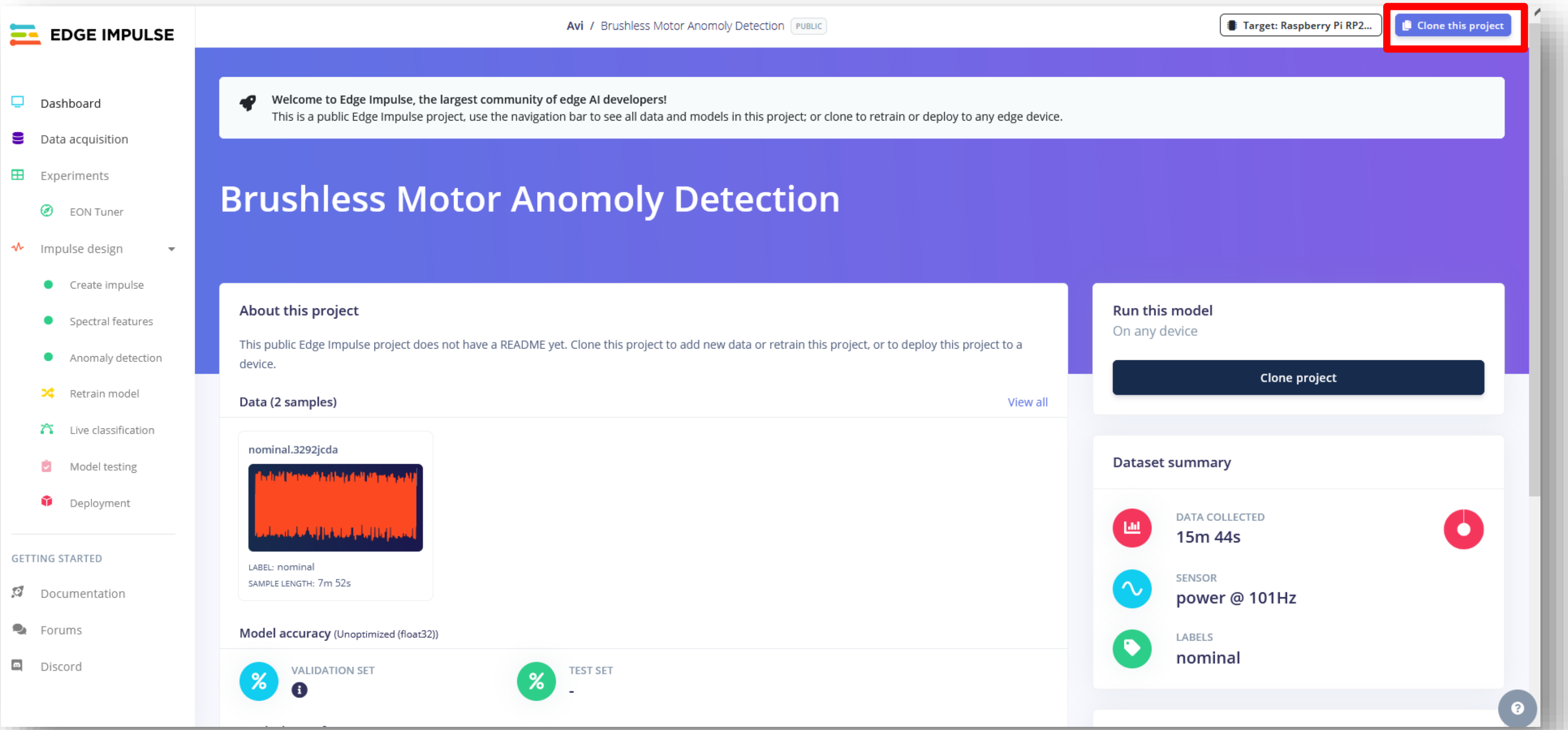


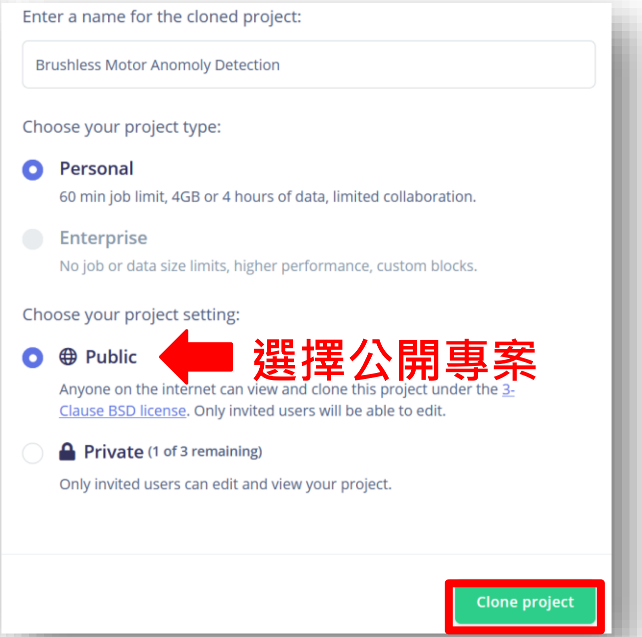


https://docs.edgeimpulse.com/projects/expert-network/brushless-dc-motor-anomaly-detection

複製並建立專案

複製專案到個人帳戶



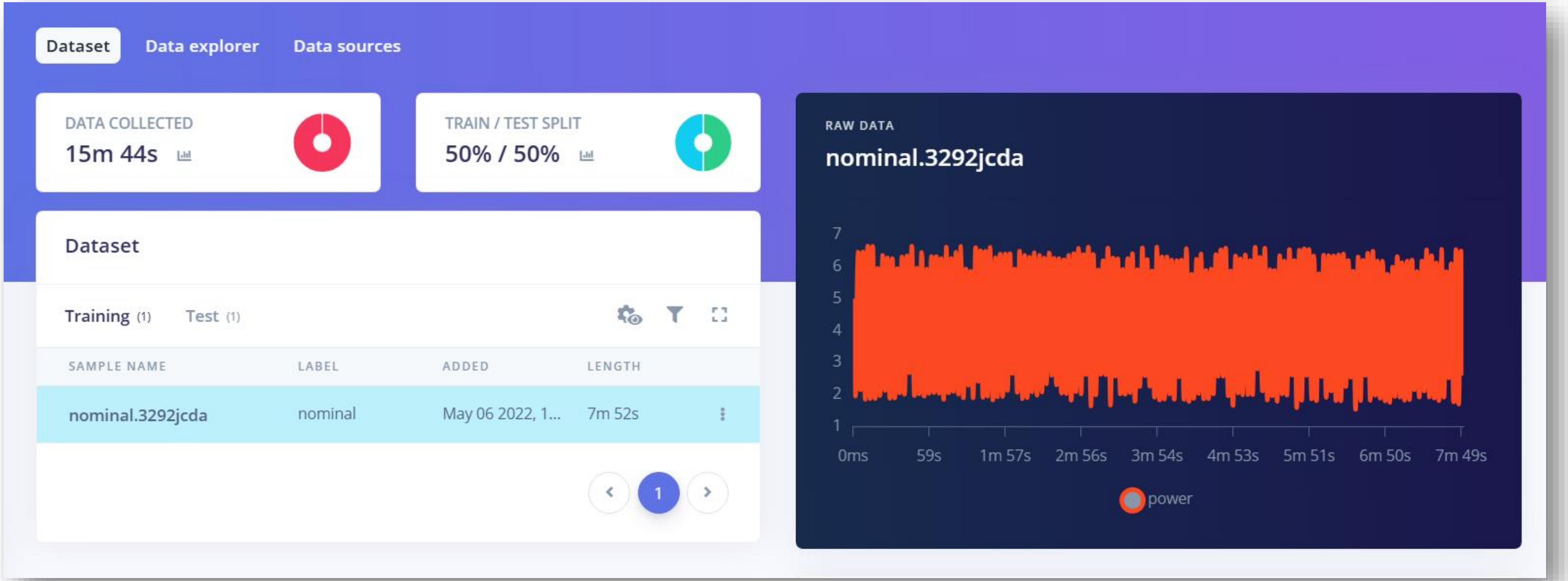


選擇公開專案

建立專案

https://studio.edgeimpulse.com/public/102584/latest

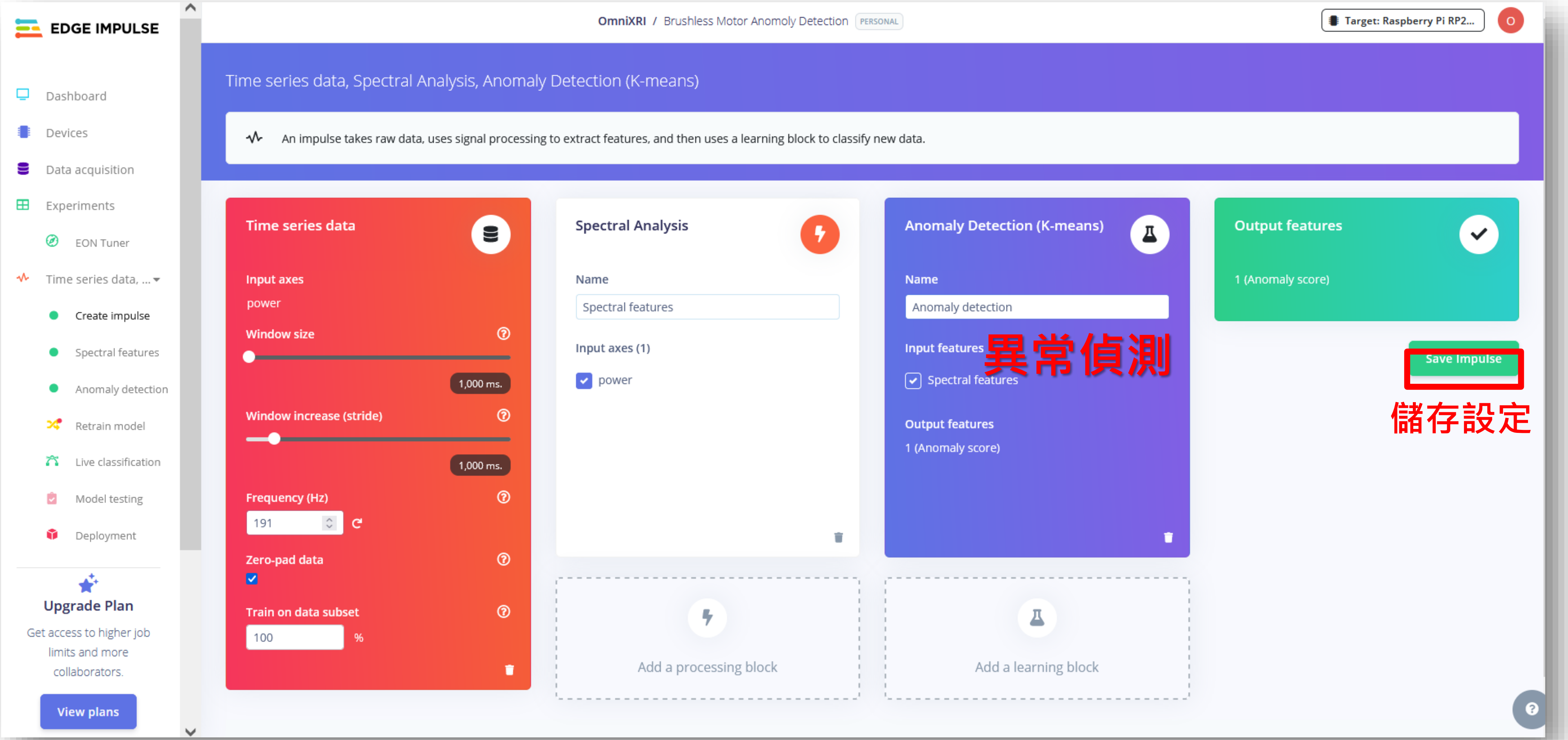
建立資料集



只需收集正常訊號不需提前分類，一筆很長的信號即可，要包含週期變化。

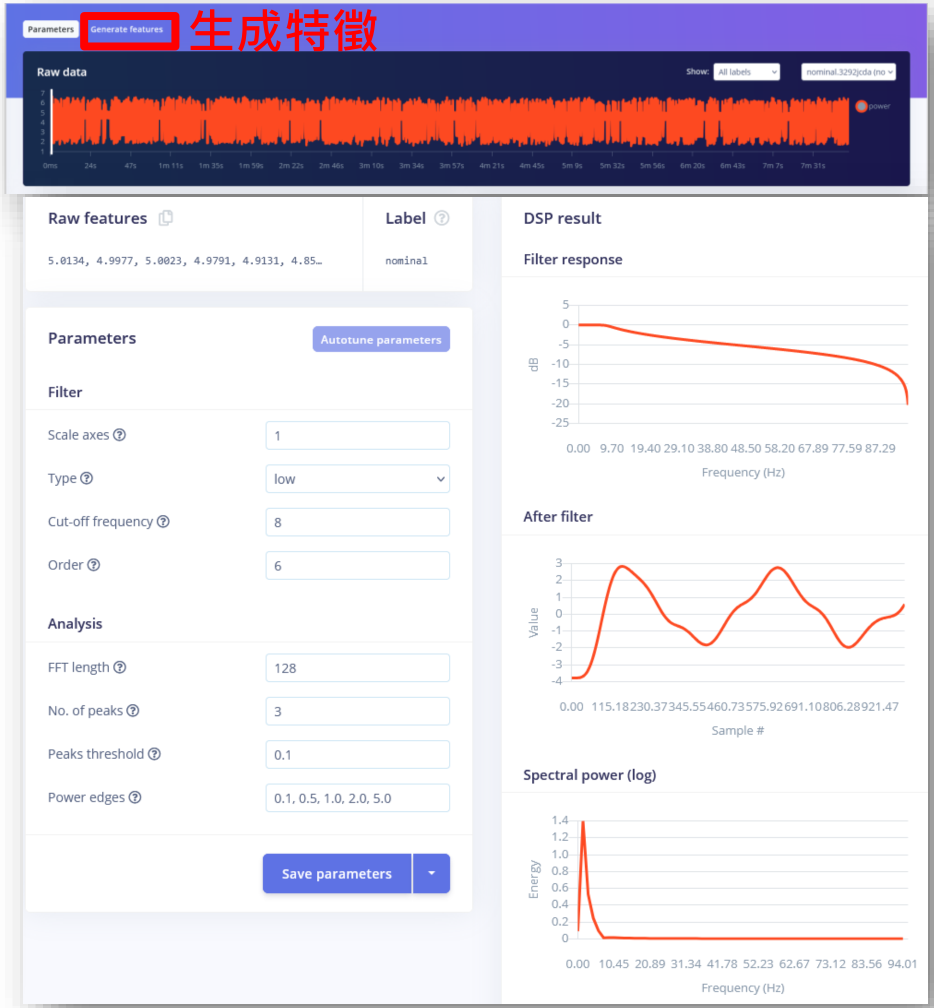
https://docs.edgeimpulse.com/projects/expert-network/brushless-dc-motor-anomaly-detection

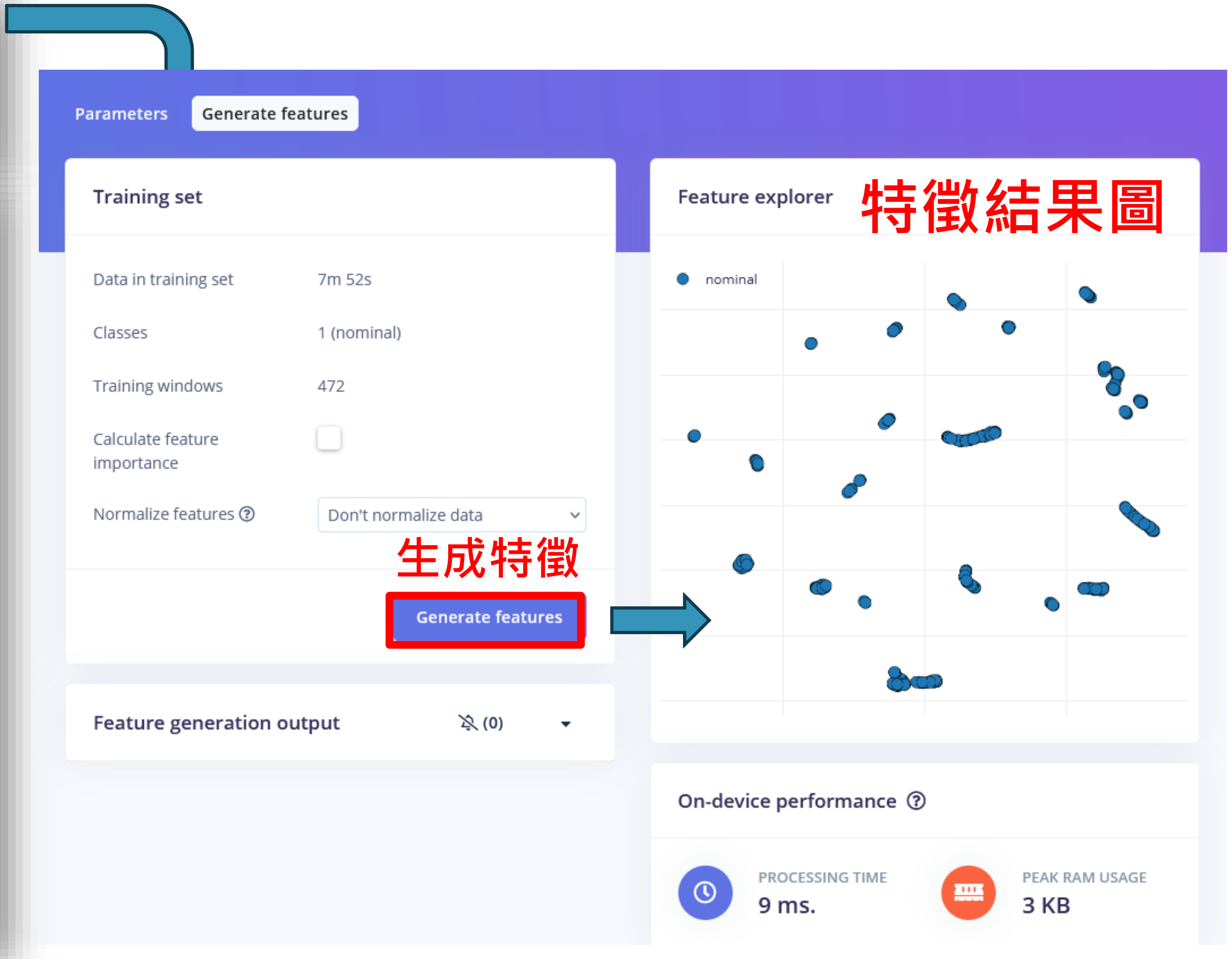
建立及設定時間序列模型



https://docs.edgeimpulse.com/projects/expert-network/brushless-dc-motor-anomaly-detection

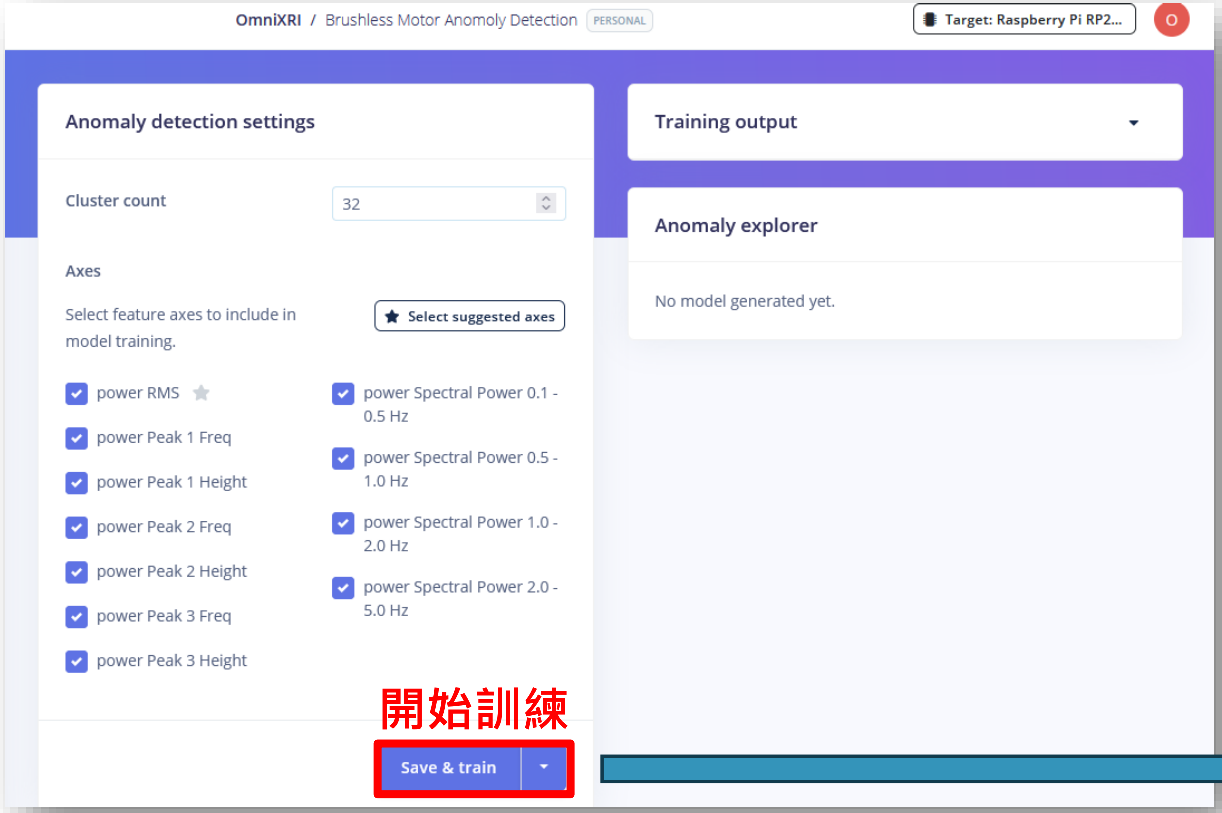
頻譜分析、特徵生成





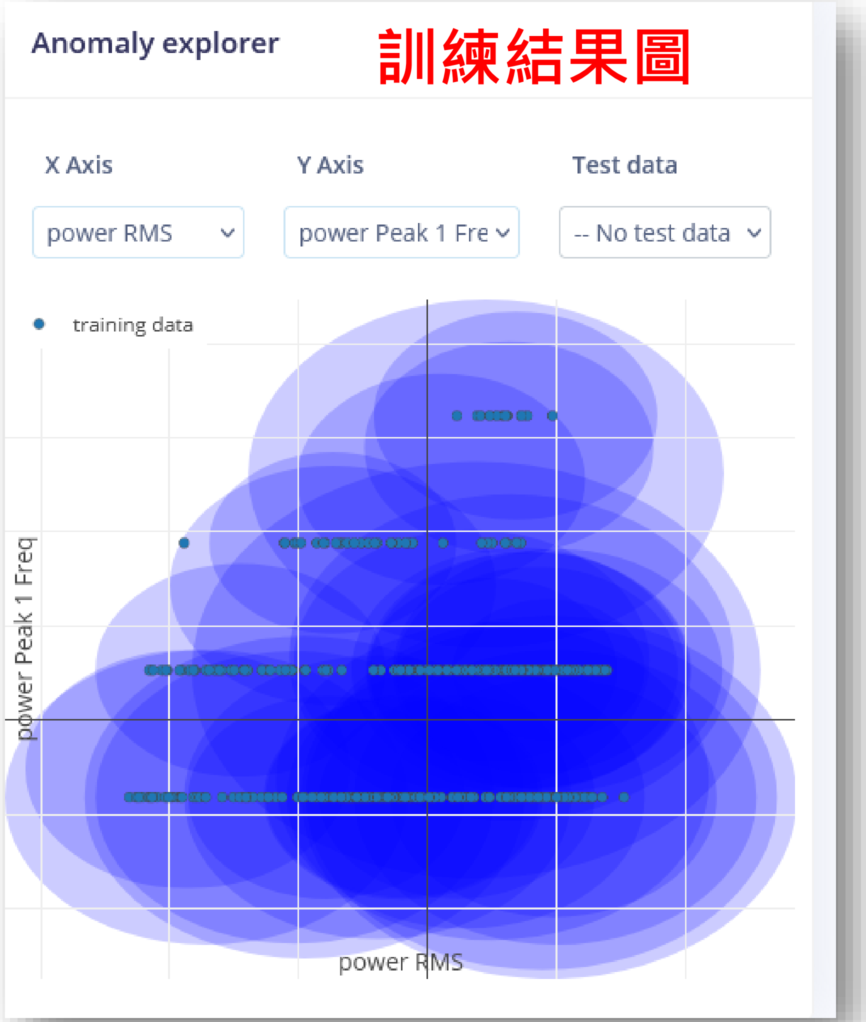
https://docs.edgeimpulse.com/projects/expert-network/brushless-dc-motor-anomaly-detection

異常偵測模型訓練



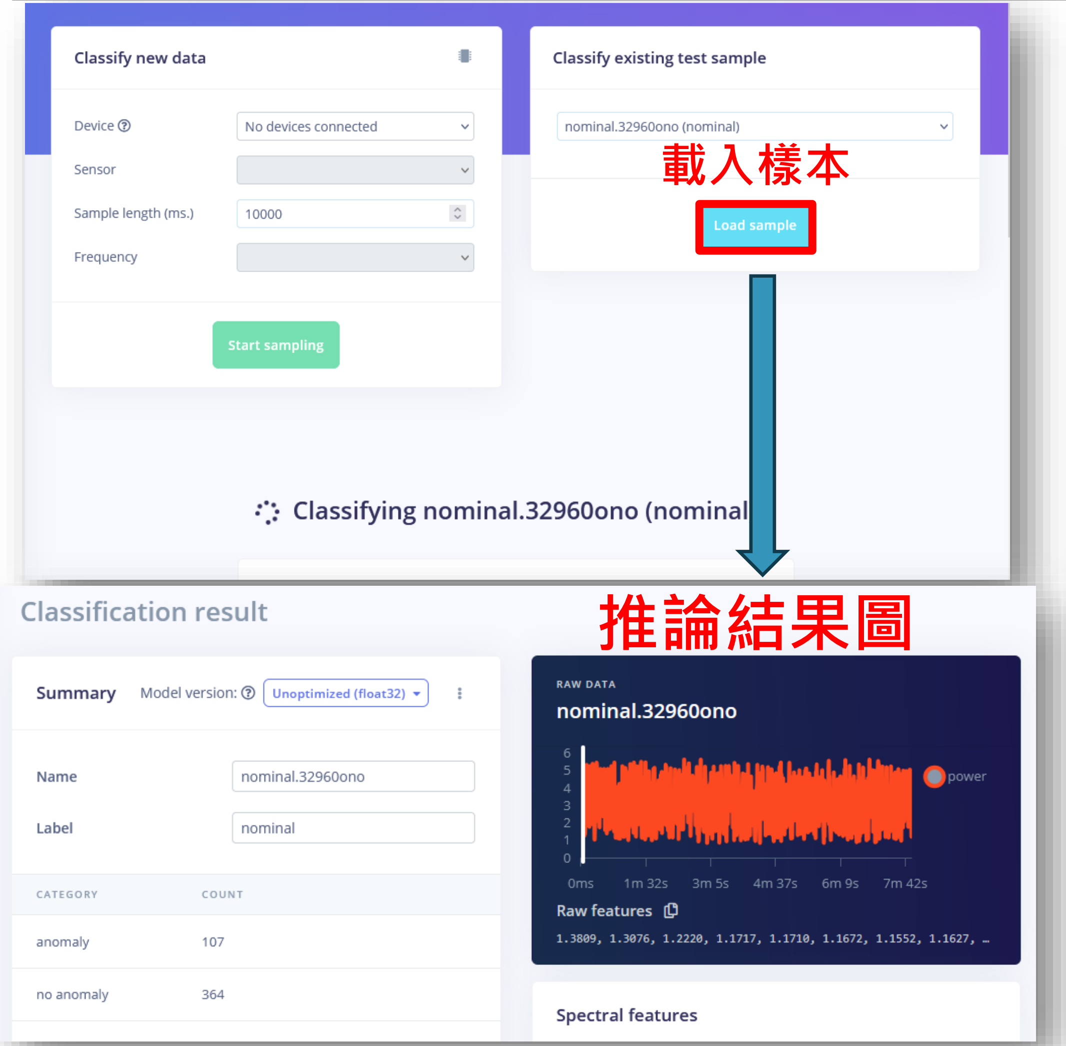
開始訓練

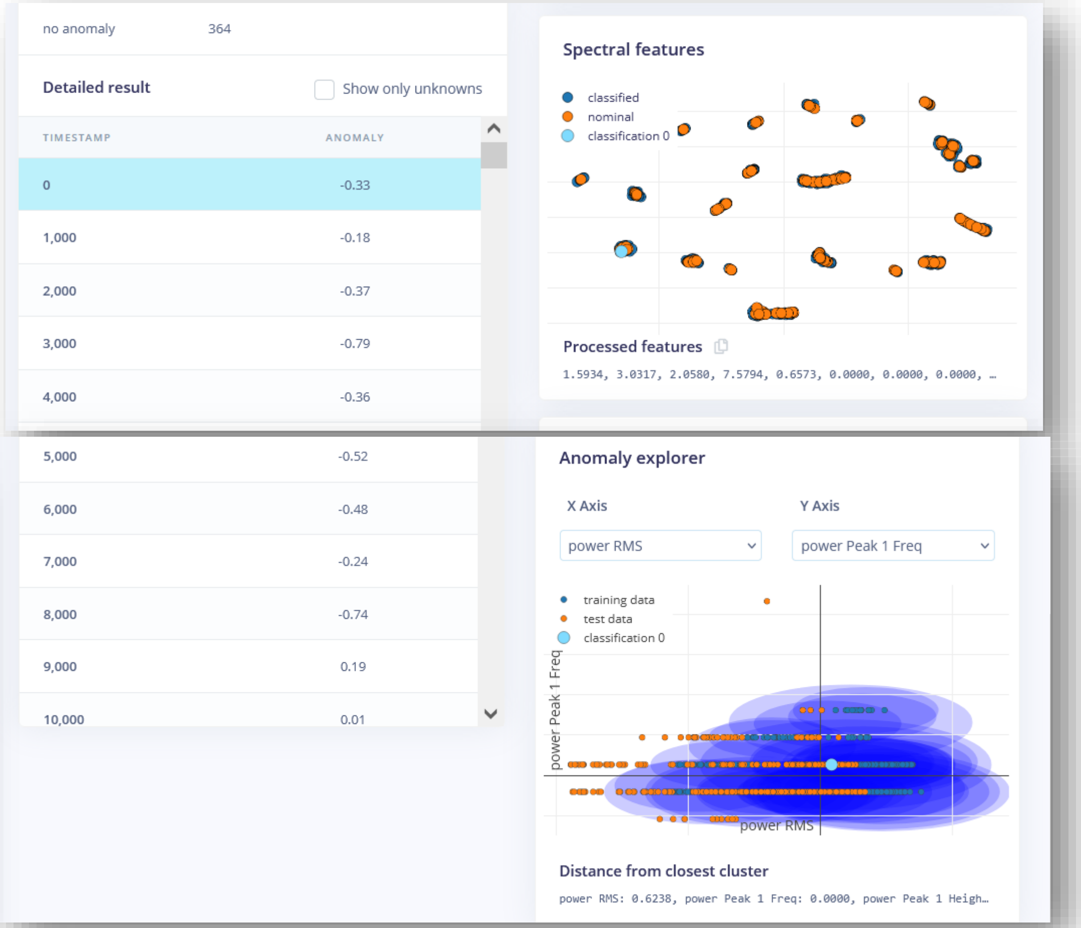
訓練結果圖



https://docs.edgeimpulse.com/projects/expert-network/brushless-dc-motor-anomaly-detection

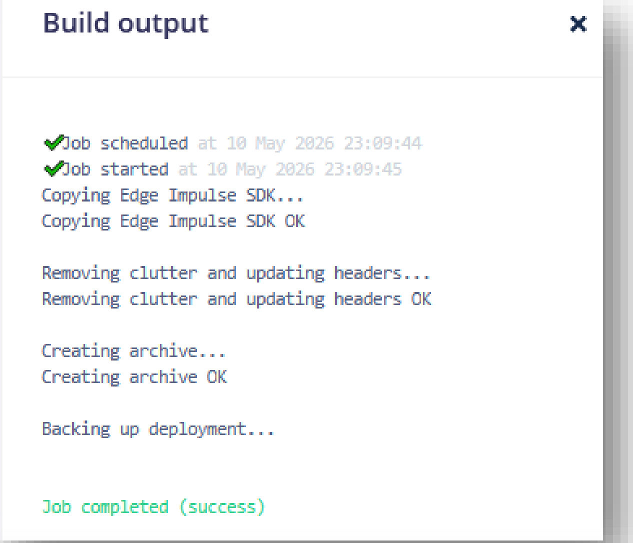
線上測試

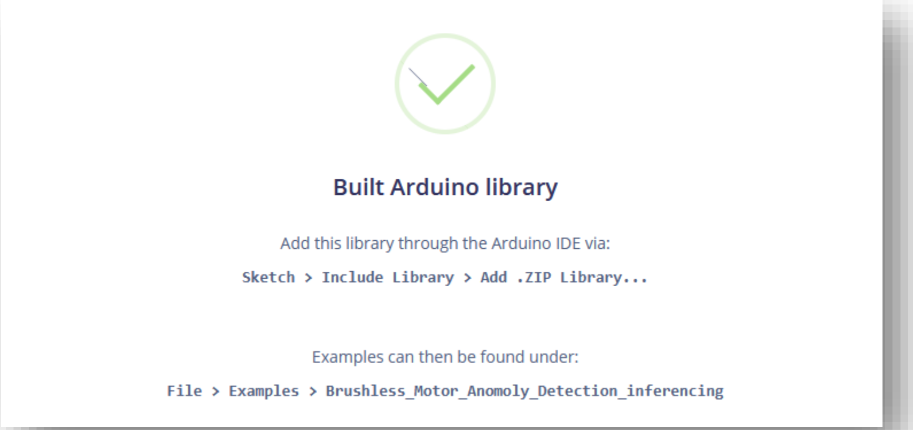




https://docs.edgeimpulse.com/projects/expert-network/brushless-dc-motor-anomaly-detection

選擇部署方式





https://docs.edgeimpulse.com/projects/expert-network/brushless-dc-motor-anomaly-detection

Arduino 端建立 Pi Pico 開發環境

1. 設定開發板管理員網址

- 開啟 Arduino IDE 2.0。
- 點選選單 **[檔案 (File)]** > **[偏好設定 (Preferences)]**。
- 在「其他開發板管理員網址 (Additional Boards Manager URLs)」欄位中，貼上以下連結：
  https://github.com/earlephilhower/arduino-pico/releases/download/global/package_rp2040_index.json
  (若已有其他網址，請用逗號隔開)
- 按下 **[確定 (OK)]**。

2. 安裝 Pico 核心

- 點選左側邊欄的 **開發板管理員圖示** (或前往 **[工具]** > **[開發板]** > **[開發板管理員...]**)。
- 在搜尋框輸入 Pico。
- 找到 **「Raspberry Pi Pico/RP2040/RP2350 by Earle F. Philhower, III」** 並點選 **[安裝 (Install)]**。

3. 導入 Edge Impulse 生成函式庫

- 開啟選單 Sketch – Include Library – **Add ZIP Library…**
- 加入剛生成的ZIP檔。
- 即可開始程式範例並執行。

https://docs.edgeimpulse.com/projects/expert-network/brushless-dc-motor-anomaly-detection

Edge Impulse Expert 案例分享 (1/2)

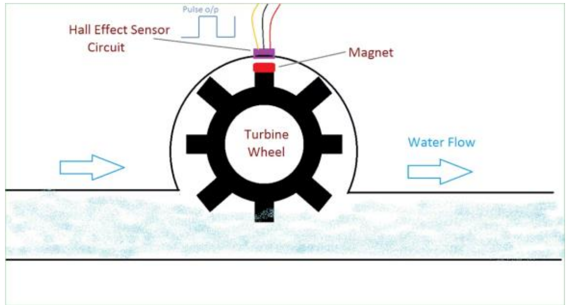
Industrial Compressor Predictive Maintenance - Nordic Thingy:53

Weight Scale Predictive Maintenance - Arduino Nano 33 BLE Sense


Anticipate Power Outages with Machine Learning - Arduino Nano 33 BLE Sense



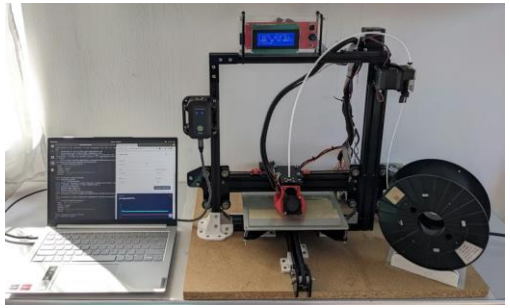
Pipeline Clog Detection with a Flowmeter and AI - Seeed Wio Terminal

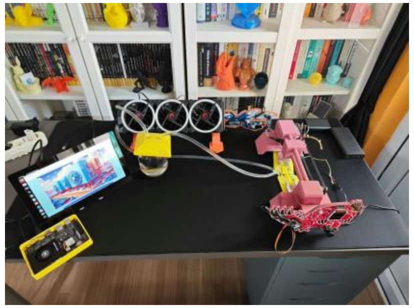
https://docs.edgeimpulse.com/projects/expert-network/project-list#predictive-maintenance-and-fault-classification-projects

Edge Impulse Expert 案例分享 (2/2)

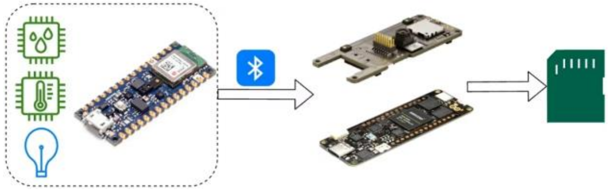
BrickML Demo Project –
3D Printer Anomaly Detection

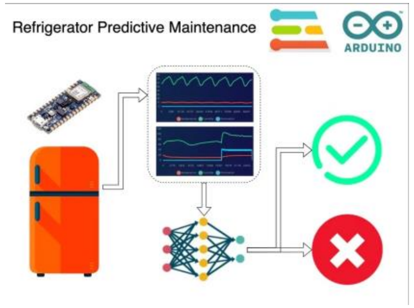


AI-driven Audio and Thermal HVAC Monitoring –
SeeedStudio XIAO ESP32

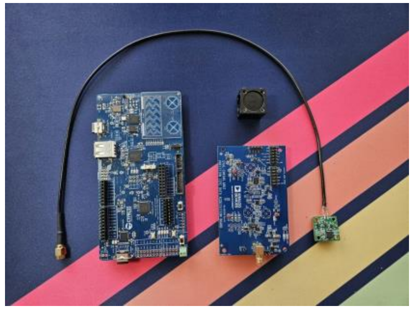


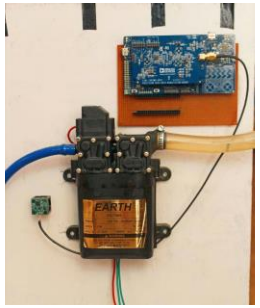
Refrigerator Predictive Maintenance –
Arduino Nano 33 BLE Sense





Motor Pump Predictive Maintenance –
Infineon PSoC 6 WiFi-BT Pioneer Kit + CN0549





https://docs.edgeimpulse.com/projects/expert-network/project-list#predictive-maintenance-and-fault-classification-projects

參考文獻

- 臺灣科技大學資訊工程系 人工智慧與邊緣運算實務 (CS5149701)
  https://omnixri.blogspot.com/p/ntust-edge-ai.html
- OmniXRI's Edge AI & TinyML 小學堂 【第13講】案例實作─運動辨識
  https://github.com/OmniXRI/Edge_AI_TinyML_Course_2024/tree/main/Ch13_Motion_Recognition
- Seeed Studio XIAO nRF52840 Sense – 硬件使用 - IMU 使用方法
  https://wiki.seeedstudio.com/cn/XIAO-BLE-Sense-IMU-Usage/
- Seeed Studio XIAO nRF52840 Sense - 嵌入式機器學習 - 基於 Edge Impulse 的動作辨識
  https://wiki.seeedstudio.com/cn/XIAOEI/
- 許哲豪，TinyML應用大全（30組案例分享）
  https://hackmd.io/@OmniXRI-Jack/tinyML_30_projects
- Edge Impulse Document – Projects – Expert Network – Accelerometer and Activity Projects
  https://docs.edgeimpulse.com/projects/expert-network/project-list#accelerometer-and-activity-projects
- Edge Impulse Document – Projects – Expert Network – Predictive Maintenance and Defect Detection Projects
  https://docs.edgeimpulse.com/projects/expert-network/project-list#predictive-maintenance-and-fault-classification-projects